The Uncertainty Premium of Climate Tipping Points [*]

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Abstract

Climate tipping points are shifts in the climate system that could lock the world into a higher-temperature regime. Many tipping points are characterised by Knightian uncertainty, that is, it is difficult to assign prior probabilities to their occurrence. This paper quantifies the economic cost of this uncertainty using a min-max regret robust policy in an integrated assessment model with temperature feedback effects. By not imposing prior probabilities over critical thresholds, the framework provides policy guidance when threshold forecasting and learning are limited. I compute the implied social cost of carbon under the robust policy and compare it to the complete information benchmark. The uncertainty premium is modest at the start of the transition, between 1% and 2.2% in 2020, but rises to approximately 20% by the end of the century as robust and complete information policy paths diverge. These results imply that tipping-point ambiguity is not a first-order driver of carbon pricing in the short run, but becomes significant over policy-relevant horizons, strengthening the case for precautionary abatement.

As the global average temperature rises due to greenhouse gas emissions from human activities, temperature feedback mechanisms in the climate system can drive the world past tipping points, into a regime of higher temperatures. Unfortunately, tipping points are very hard (Ditlevsen and Ditlevsen, 2023; Ditlevsen and Johnsen, 2010; Rietkerk et al., 2025; Smolders, van Westen and Dijkstra, 2024), if not impossible (Ben-Yami et al., 2024), to predict, giving policymakers little to no time to react once they are crossed.

In this paper, I study the economic costs of Knightian uncertainty around climate tipping points. I address this uncertainty non-probabilistically by defining and computing a tipping-point uncertainty premium on the social cost of carbon. Using an integrated assessment model that incorporates temperature feedback effects, I first compute optimal emissions abatement strategies across multiple tipping-point scenarios calibrated to reflect recent evidence in the climate literature (Armstrong McKay et al., 2022; Deutloff, Held and Lenton, 2025; Seaver Wang et al., 2023). I then construct a min-max regret robust policy over a set of plausible tipping thresholds and compute the associated social cost of carbon. I interpret the relative difference between this robust social cost of carbon and the complete information benchmark as an uncertainty premium.

I show that the uncertainty premium on the social cost of carbon is modest at the start of the transition, between 1% and 2.2% in 2020, but rises to approximately 20% by 2100. This increase emerges under a robust policy that hedges across tipping-point scenarios by balancing two sources of regret: under-abating when the threshold is low and imminent, and over-abating when the threshold is remote and non-binding. Quantitatively, the robust policy places substantial weight on both low-threshold and high-threshold cases, implying stronger abatement than in a climate without tipping but weaker abatement than under a known imminent tipping point. As a result, robust and complete information policy paths are initially close, generating only a limited short-run wedge in carbon pricing. Over time, however, the implied trajectories of emissions, carbon concentration, and temperature increasingly diverge, so the welfare cost of unresolved threshold ambiguity compounds. This uncertainty premium therefore captures not only contemporaneous ambiguity, but also the legacy of past robust abatement choices made without full information. These findings imply that tipping-point uncertainty is not a first-order driver of the social cost of carbon in the short run, yet becomes significant over policy-relevant horizons, reinforcing the case for a precautionary orientation in carbon-pricing policy.

The rest of the paper is structured as follows. Section 1 puts the current paper in the context of the climate and environmental economic literature and highlights its contributions.

Section 2 introduces the main components of the model: the climate, the economy, and the social planner problem. Section 3 computes optimal emissions abatement policies in a complete information benchmark, that is, when tipping points are known. Section 4 develops the robust policy under threshold uncertainty and quantifies the associated uncertainty premium on the social cost of carbon. Finally, Section 5 draws policy implications and concludes.

1 Related Literature

A large body of recent economic literature has highlighted the importance of correctly incorporating climate dynamics when analysing the economic trade-off between current emissions and the reduction of future climate damages induced by rising temperatures. The key challenge is to develop integrated assessment models that, on the one hand, are sufficiently simple to be integrated into a dynamic optimisation problem, necessary to compute the economic costs of climate change and optimal emission abatement policies, and, on the other hand, can reproduce the key dynamics of larger, more accurate climate models (Dietz et al., 2021*a*). One dynamic aspect of the climate that has drawn much attention in economics is the concept of climate tipping points (Li, Crépin and Lindahl, 2024).

Early work on uncertain tipping points in environmental economics has modelled them using stochastic jump processes, building on the work on economic catastrophes by Barro (2009). At each moment in time, either temperature, atmospheric CO_2 concentration or climate damages to consumption experience a discontinuous jump with some probability known to the planner. This approach was introduced to climate economics by Pindyck and Wang (2013)[1]. It has been used in modelling climate tipping points by, for example, Lontzek et al. (2015), and Van der Ploeg and De Zeeuw (2018). The broad adoption of this approach is due to its flexibility as it allows for analytical tractability (Li, Crépin and Lindahl, 2024; Lin and Van Wijnbergen, 2023; Van den Bremer and Van der Ploeg, 2021), the introduction of uncertainty and ambiguity aversion (Olijslagers and Van Wijnbergen, 2019), and calibration (Hambel, Kraft and Schwartz, 2021). Nevertheless, this modelling choice neglects two crucial features of climate tipping points that are the focus of this paper. First, crossing a tipping point can push the climate into a new, persistent regime of high temperatures. Barro disasters are not suited to model these different regimes, as they model only transient catastrophic events. Second, it requires defining a probability distribution over the tipping event, whether

[1] This approach has an earlier history in the optimal control of environmental systems (Kamien and Schwartz, 1971; Nævdal and Oppenheimer, 2007; Tsur and Zemel, 1996, among others). I refer to Li, Crépin and Lindahl (2024, Section 3) for a detailed review.

real or the social planner's belief, which is, in reality, hard to compute reliably.

A second approach is to model feedback in climate dynamics explicitly. This is closer to the real behaviour of the climate system (McGuffie and Henderson-Sellers, 2014), as it introduces multiple climate regimes. This is the standard approach adopted in more complex climate modelling (e.g. Smith et al. 2017). Furthermore, optimisation over such systems has a long history in economics (Skiba, 1978) and has been developed extensively in environmental (Mäler, Xepapadeas and de Zeeuw, 2003; Wagener, 2013) and climate economics (Greiner and Semmler, 2005; Nordhaus, 2019; Wagener, 2015). Modelling the feedback explicitly has one drawback: it requires assuming the size of the feedback and when it comes into effect. These two properties are objects of empirical contention (Armstrong McKay et al., 2022; Seaver Wang et al., 2023). Furthermore, computationally solving such models can be technically challenging. Because of this, models of climate change with positive feedback are less commonly used when attempting to provide quantitative economic estimates via calibration.[2]

A third approach is to assume that with some state dependent probability either the marginal damages of temperature (Cai and Lontzek, 2019; Cai, Lenton and Lontzek, 2016) or the mapping between carbon concentration and temperature (Lemoine and Traeger, 2016*a*,*b*, 2014) change, such that the system dynamics follow a Markov chain. Among these, Lemoine and Traeger (2014), Lemoine and Traeger (2016*a*) and the present paper deal with the uncertainty around the tipping point. The former assumes a Bayesian planner has a prior distribution over the tipping point and updates their belief as climate information arises. This assumption allows the authors to construct policies that deal with the tipping point uncertainty and react over time. In the context of climate tipping points, this approach suffers from two limitations. First, it requires an assumption about the distribution of the tipping point, or a prior over it. Lemoine and Traeger (2016*a*, 2014) take this to be uniform. While possibly justified by the principle of insufficient reason, a uniform prior over the tipping threshold implicitly induces informative distributions over outcome variables central to the decision problem, such as climate damages and tipping probabilities (Gilboa, 2009). Second, in this literature, it is assumed that the planner can incorporate information from the climate and update their priors. This assumption might not be consistent with recent climate literature (Ben-Yami et al., 2024; Ditlevsen and Ditlevsen, 2023) and possibly practically infeasible for many fast-moving systems (Rietkerk et al., 2025). To deal with these limitations, Lemoine

[2] As above, in this short paragraph, I have only focused on a few papers from this literature. I again refer the reader to Li, Crépin and Lindahl (2024, Section 4) for a more comprehensive review.

and Traeger (2016*a*) extend Lemoine and Traeger (2014), addressing the Knightian uncertainty around tipping points by deriving a premium on the social cost of carbon from smooth ambiguity preferences. Nevertheless, this probabilistic approach requires an exogenous hazard rate on the tipping event to be ex-ante defined. The present paper complements Lemoine and Traeger (2016*a*) by developing a non-probabilistic tipping point uncertainty premium.

Lastly, a recent stream of literature (most notably Dietz et al. 2021*b*) circumvents the issue by foregoing the computation of optimal policies and embedding a simple and exogenous economic module into a complex climate model. Quantities of interest are then computed via Monte Carlo experiments. This approach allows economists to compute economic outcomes of climate tipping points by integrating state-of-the-art climate science, at the cost of not being able to analyse counterfactual policies.

In this paper, I consider a tipping point induced by positive feedback in the temperature dynamics. I calibrate the feedback to match temperature dynamics in recent climate modelling of tipping points. I treat the Knightian uncertainty around the tipping point non-probabilistically and develop a novel measure of uncertainty premium in the social cost of carbon. The computation of the uncertainty premium builds on a long literature on the use of decision-theory rules, both probabilistic and non-probabilistic, in decision-making under Knightian uncertainty in climate economics (Athanassoglou and Xepapadeas, 2012; Heal and Millner, 2014; Millner, Dietz and Heal, 2013; Rudik, 2020, among others). When dealing with Knightian uncertainty, there is a trade-off between a more informationally demanding probabilistic approach and a less information-demanding non-probabilistic approach (Heal and Millner, 2014). The former has been successfully employed to study tipping point uncertainty in climate economics (Lemoine and Traeger, 2016*a*; Traeger, 2014). The non-probabilistic approach taken here is then complementary to that of Lemoine and Traeger (2016*a*). On the one hand, it gives a more robust policy prescription, that is, a conservative and non-probabilistic uncertainty premium on the optimal carbon tax. On the other hand, the policy prescription is less rich as it does not use possible prior information the planner might have on the tipping point. As such information might be unreliable, this complementary view aligns with Heal and Millner (2014), who argues that climate policy must be grounded in aversion to risk and ambiguity, accounting for the small but positive risk of disastrous outcomes.

Finally, the numerical solver employed in the paper contributes to the literature on controlled stochastic processes by developing a parallelisation algorithm, based on Bierkens, Fearnhead and Roberts (2019), for the class of solvers introduced in Kushner and Dupuis (2001), and extending their results to recursive utilities.

2 Model

The following describes the climate and economic components of the model. The model builds on the formulation proposed in Hambel, Kraft and Schwartz (2021). First, Section 2.1.1 introduces carbon sinks and their interaction with atmospheric carbon concentration. Then, Section 2.1.2 defines the temperature dynamics and the temperature feedback mechanism that leads to a tipping point. Section 2.1.3 discusses the resulting tipping point in the climate system. Finally, Section 2.2 presents the economy.

2.1 Climate

2.1.1 Carbon sinks and $\mathrm{CO_2e}$ concentration

Emissions from human economic activity E_t increase the atmospheric concentration M_t of greenhouse gasses (GHGs), measured in their $\mathrm{CO_2}$ equivalent, hereafter $\mathrm{CO_2e}$. Part of this, in turn, decays into natural sinks. I denote by $\delta_m(M_t)$ the rate of decay of $\mathrm{CO_2e}$ concentration M_t into natural sinks. The function δ_m is decreasing in the atmospheric carbon concentration. This assumption is in line with the empirical evidence and reflects the fact that it is much easier to stabilise atmospheric $\mathrm{CO_2e}$ concentration M_t if natural carbon sinks are not saturated, that is, for low values of M_t (Le Quéré et al., 2007; Shi et al., 2021).

In absence of abatement efforts, atmospheric $\mathrm{CO_2e}$ concentration M_t evolves according to the *no-policy* SSP5-8.5 scenario (Gidden et al., 2019). Variables under this scenario are indexed by the superscript np: let E_t^{np} be the no-policy emissions, and M_t^{np} the resulting atmospheric $\mathrm{CO_2e}$ concentration. No-policy atmospheric $\mathrm{CO_2e}$ concentration M_t^{np} evolves as

$$\frac{\mathrm{d}M_t^{\mathrm{np}}}{M_t^{\mathrm{np}}} = \gamma_t^{\mathrm{np}}\mathrm{d}t + \sigma_m \mathrm{d}W_{m,t}, \tag{1}$$

where the no-policy expected growth rate of atmospheric $\mathrm{CO_2e}$ concentration is given by

$$\gamma_t^{\mathrm{np}} := \frac{E_t^{\mathrm{np}}}{M_t^{\mathrm{np}}} - \delta_m(M_t^{\mathrm{np}}) \tag{2}$$

and $W_{m,t}$ is a standard Brownian motion. The Brownian motion accounts for possible uncontrolled and unexpected shocks to $\mathrm{CO_2e}$ concentration, such as fluctuations in the absorption capacity of carbon sinks (DeVries et al., 2019). The variance σ_m^2 of these shocks is assumed to be constant. The no-policy scenario describes an energy intensive future, in which fossil fuel usage develops rapidly and little to no abatement takes place. I calibrate the implied growth rate of CO_2e concentration γ_t^{np}, using the projected CO_2e concentration path M_t^{np} from Gidden et al. (2019). The calibration is described in B.1.1 and its results are shown in Figure 1. The left figure shows the path of the no-policy growth rate γ_t^{np} (black solid line with point

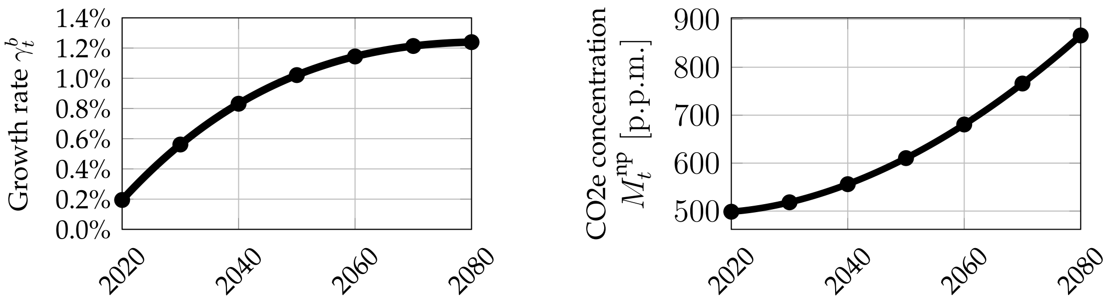

Figure 1: Growth rate of CO_2e concentration in the no-policy scenario γ_t^{np} (black solid line with point markers) and median path of no-policy CO_2e concentration M_t^{np} (black solid line) from (1).

markers) and the right figure shows the resulting growth of CO_2e concentration M_t^{np} (black solid line). The CO_2e concentration in this scenario grows at an increasingly faster rate until approximately 2070, when the growth rate peaks just below 1.4%. Thereafter, the growth rate starts declining.

Abatement efforts α_t lower the growth rate of CO_2e concentration M_t vis-à-vis the no-policy scenario M_t^{np}. That is, the growth rate of CO_2e concentration M_t satisfies

$$\frac{\mathrm{d}M_t}{M_t} = \left(\gamma_t^{\mathrm{np}} - \alpha_t\right)\mathrm{d}t + \sigma_m \mathrm{d}W_{m,t}. \tag{3}$$

Implementing no abatement policy $\alpha_t = 0$ corresponds to a no-policy scenario $M_t = M_t^{\mathrm{np}}$, while implementing a full abatement policy $\alpha_t = \gamma_t^{\mathrm{np}}$ stabilises CO_2e concentration. Any abatement policy α_t can be linked to the implied level of emissions by introducing the emission reduction rate $\varepsilon(\alpha_t)$, which keeps tracks of what fraction of emissions is being abated

$$E_t = \left(1 - \varepsilon(\alpha_t)\right) E_t^{\mathrm{np}}. \tag{4}$$

Throughout the paper, policy results are reported in the faction of emissions abated $\varepsilon_t = \varepsilon(\alpha_t)$ rather than the abatement rate α_t. If abatement α_t exceeds the growth rate γ_t^{np} of CO_2e and the decay in carbon syncs $\delta_m(M_t)$, that is,

$$\alpha_t > \gamma_t^{\mathrm{np}} + \delta_m(M_t), \tag{5}$$

greenhouse gasses are sequestered from the atmosphere and atmospheric concentration M_t decreases. In this case, the fraction of abated emissions is more than 100%, that is, $\varepsilon_t > 1$.

2.1.2 Temperature

Average world temperature T_t $[°]$ in deviation from pre-industrial levels is determined by a zero-dimensional energy balance model. Earth's energy balance, in its simplest form, prescribes that temperature tends towards balancing incoming solar radiation S and outgoing long-wave radiations $\eta\sigma T_t^4$, where σ is the Stefan-Boltzmann constant and η is an emissivity rate. Due to the presence of GHGs, certain wavelengths are scattered and not radiated outwards (Ghil and Childress, 2012). This introduces an additional radiative forcing G which yields the balance equation $S = \eta\sigma T_t^4 - G$. The GHGs radiative forcing term G can be decomposed into a constant component G_0 and a component which depends on the equilibrium level of $\mathrm{CO_2e}$ concentration in the atmosphere M_t relative to the pre-industrial level M^p, such that

$$G = G_0 + G_1 \log(M_t/M^p). \tag{6}$$

The parameters G_0 and G_1 are calibrated to match the mapping between GHGs and temperature deviations in the FaIRv2 model (Leach et al., 2021). The calibration procedure is described in B.1.3. This zero-dimensional climate model does not capture the heterogeneity of temperature deviations and, as a consequence, climate damages, across the globe (McGuffie and Henderson-Sellers, 2014). Another limitation is that the temperature dynamics and the carbon cycle model are simpler than those commonly employed in the literature. Despite this, when targeting the average world temperature, the calibration presented here closely tracks the FaIRv2 projection (Leach et al., 2021) both in- and out-of-sample (see Figure 18 in B.1.3). Importantly, it closely tracks the timing of the temperature impulse response to an increase in atmospheric $\mathrm{CO_2e}$ concentration, which is a known limitation of climate modules in IAMs (Dietz et al., 2021*a*). I choose to match FaIRv2 projection as they are used in Deutloff, Held and Lenton (2025), which then serves as the basis for the tipping point calibration in this paper. Furthermore, the simple temperature dynamics allow us to solve for optimal closed-loop abatement policies, which are used to compute richer counterfactuals, as in Hambel, Kraft and Schwartz (2021). For an alternative approach to the study of tipping points, with a richer climate model but without optimal abatement policies, I refer the reader to Dietz et al. (2021*b*).

The climate system presents mechanisms that act as positive temperature feedbacks: an increase in global mean surface temperature can lead to additional forcing, which, in turn, further increases temperatures. For some of these mechanisms, the feedback is sufficiently strong that after a critical temperature level, referred to as a climate tipping point, the feedback becomes self-reinforcing and temperature increases to a hotter regime, even in the absence of further GHG emissions from human activity. Such mechanisms are commonly referred to as tipping elements. Armstrong McKay et al. (2022) have identified nine salient global tipping elements in the context of climate change. The tipping elements which can potentially lead to the most significant warming are an abrupt permafrost collapse and the Amazon forest dieback. These tipping elements lead to an increase in emissions and, hence, radiative forcing as temperature rise (Parry, Ritchie and Cox, 2022; Turetsky et al., 2019). In this paper, I jointly model these tipping elements as an additional forcing factor $\lambda^{T^c}(T_t)$ which is increasing in temperature T_t (McGuffie and Henderson-Sellers, 2014). The function $\lambda^{T^c}(T_t)$ transitions from no additional forcing to an additional forcing level ΔS via a smooth transition function $L(T_t - T^c)$, that is,

$$\lambda^{T^c}(T_t) \coloneqq \Delta S \; L(T_t - T^c), \tag{7}$$

where T^c $[°]$ is the critical temperature at which the feedback becomes self-reinforcing. The function L and the additional forcing ΔS are calibrated, in B.1.4, to match the additional warming caused by tipping elements in Deutloff, Held and Lenton (2025). An important distinction between the feedback mechanism presented here and that of Deutloff, Held and Lenton (2025) is that, in the latter, each tipping element is modelled as an additional positive mechanism that contributes to GHG emissions. This approach is more realistic than that taken in this paper and allows the authors to study interactions among various tipping elements. Nevertheless, the stylised formulation provided here is analytically tractable, while still capturing two key features of the results of Deutloff, Held and Lenton (2025). First, it captures the timing: once a tipping point is triggered, the additional warming is, on average, fully realised in 10.5 (± 4.64) years. Second, it captures the magnitude, as the average additional warming peaks at 0.8 $(\pm 0.3)°$.

The only parameter which is not calibrated is the threshold temperature T^c[3]. Best estimates from climate sciences are that most relevant transitions occur for temperatures T_t between $2°$ and $5°$ over pre-industrial levels (Armstrong McKay et al., 2022; Seaver Wang et al., 2023). A large body of literature has focused on estimating critical thresholds associated with climate tipping points (e.g., Boulton, Allison and Lenton 2014; Van Westen, Kliphuis and Dijkstra 2024). However, for many climate feedback mechanisms, this can be very hard, if not impossible, to do (Ben-Yami et al., 2024; Ditlevsen and Ditlevsen, 2023; Ditlevsen and

[3] In the simulations by Deutloff, Held and Lenton (2025), the authors sample uniformly on the possible critical threshold ranges provided in Armstrong McKay et al. (2022)

Johnsen, 2010; Rietkerk et al., 2025; Wagener, 2013). In line with this, to avoid attaching an arbitrary prior over the critical threshold T^c, in this paper I assume the uncertainty in T^c to be Knightian. Figure 2 shows the calibrated transition function (7) under an imminent tipping scenario $T^c = 2°$ (black line, square markers), a remote tipping scenario $T^c = 3°$ (dark grey line, diamond markers), and the linear model (light grey line, circle marker). Despite being

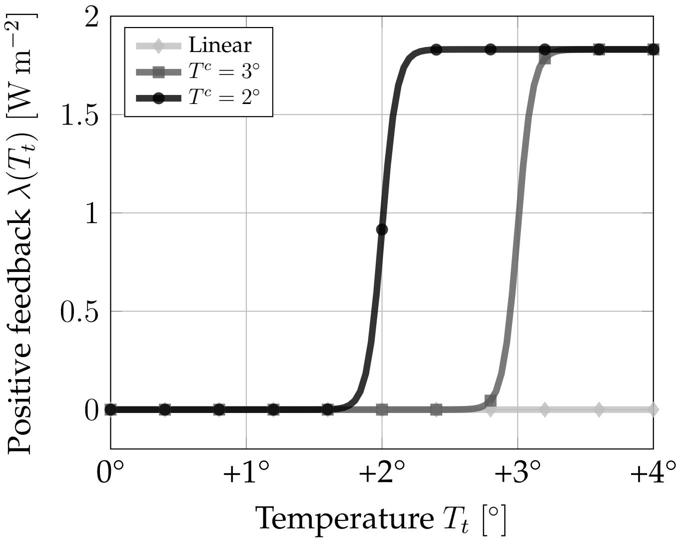

Figure 2: Temperature feedback $\lambda^{T^c}(T_t)$ in an imminent tipping scenario $T^c = 2°$ (black line, square markers), a remote tipping scenario $T^c = 3°$ (dark grey line, diamond markers), and the linear model (light grey line, circle marker).

a highly stylised and reduced form representation of a complex and spatially heterogeneous process, λ^{T^c} captures the core mechanism behind feedback processes in the temperature dynamics that can generate tipping points (McGuffie and Henderson-Sellers, 2014). This simple temperature feedback allows us to discuss and estimate the costs of transitioning to a new regime, what ought to be (or not be) done to prevent this from happening, and the cost associated with the uncertainty around the tipping point.

Putting these processes together we can write down the two determinants of temperature dynamics: radiative forcing, which only depends on temperature,

$$r(T_t) := S_0 + \lambda^{T^c}(T_t) - \eta\sigma T_t^4 \tag{8}$$

and the greenhouse gas effect, which only depends on the deviation of atmospheric $\mathrm{CO_2e}$ concentration to its pre-industrial level,

$$g(M_t) := G_0 + G_1 \log(M_t/M^p). \tag{9}$$

Under these two drivers, temperature change satisfies

$$\epsilon \mathrm{d}T_t = r^{T^c}(T_t)\mathrm{d}t + g(M_t)\mathrm{d}t + \sigma_T \, \mathrm{d}W_{T,t}, \tag{10}$$

where ϵ is Earth's total heat capacity and $W_{T,t}$ is a Brownian motion, modelling uncontrolled shocks to temperature, which are assumed to have constant variance σ_T^2 and be uncorrelated with shocks $W_{m,t}$ to atmospheric $\mathrm{CO_2e}$ concentration.

2.1.3 Temperature Dynamics with a Tipping Point

The presence of the temperature feedback λ^{T^c} introduces a tipping point. In the following, I discuss the consequences of this tipping point for the temperature dynamics.

For a given level of $\mathrm{CO_2e}$ concentration M_t, the temperature T_t tends towards an equilibrium $\bar{T}$ that gives radiative balance in the temperature dynamics (10), that is,

$$r^{T^c}(\bar{T}) = -g(M_t). \tag{11}$$

In the presence of a positive feedback λ^{T^c}, for some levels of atmospheric $\mathrm{CO_2e}$ concentration M_t there are multiple levels of temperature $\bar{T}$ which are in radiative balance. That is, there are multiple solutions to (11). More details on the determinants of these levels of concentration are provided in A. At these levels of atmospheric $\mathrm{CO_2e}$ concentration M_t, a positive shock in temperature is amplified by the temperature feedback λ^T. If this process becomes self-reinforcing, the temperature is pushed upwards until the temperature feedback effects flats out. This is illustrated in Figure 3. The Figure shows equilibria of temperature as a function of $\mathrm{CO_2e}$ concentration when $T^c = 2°$ (black solid line, square markers), when $T^c = 3°$ (dark grey solid line, diamond markers), and when there is no temperature feedback λ^{T^c} (light grey solid line, circle markers). The dotted branches are unstable in the two tipping scenarios. Without temperature feedback λ^{T^c}, the equilibrium temperature is unique, and it rises with the logarithm of $\mathrm{CO_2e}$ concentration. If a temperature feedback λ^{T^c} is introduced, for some level $\mathrm{CO_2e}$ concentration, two possible temperature regimes can be sustained: a low and a high temperature one. These are illustrated as dotted lines. In the $T^c = 2°$ scenario, for a $\mathrm{CO_2e}$ concentration M_t approximately at 550 p.p.m. both a low temperature regime, with $T_t < 2°$, and a high temperature regime, with $T_t > 2°$ are possible. In the $T^c = 3°$ scenario this phenomenon occurs only when $\mathrm{CO_2e}$ concentration M_t is approximately 725 p.p.m.. In a low temperature regime, the presence of a high temperature regime is hard to detect as the

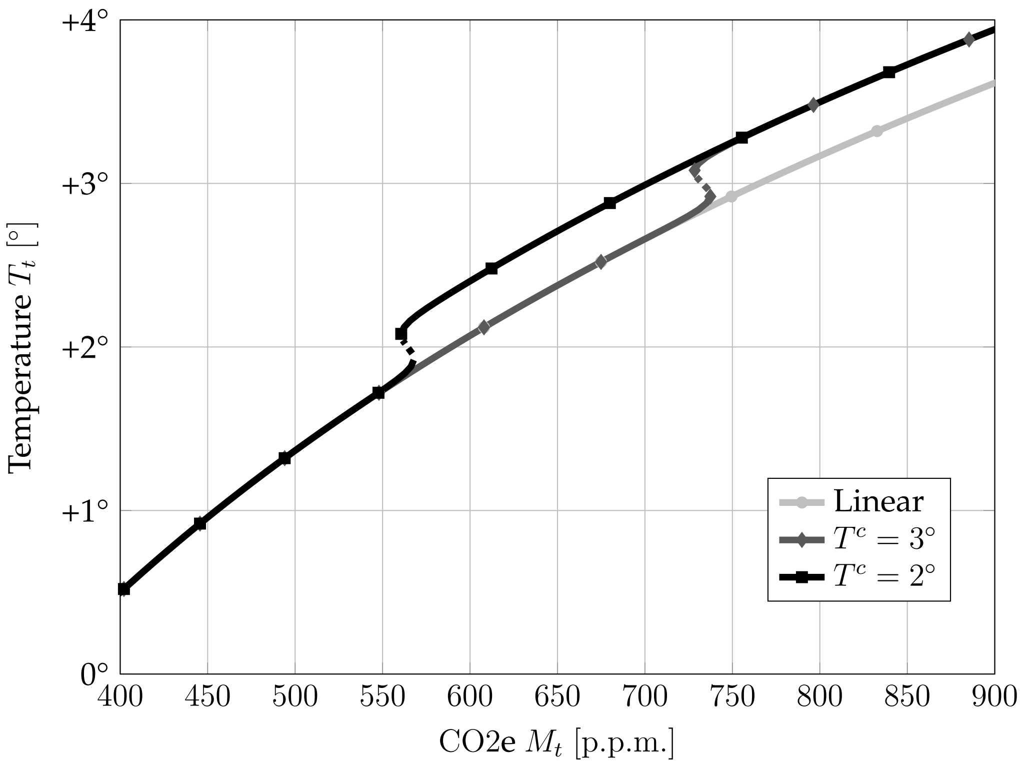

Figure 3: Equilibria of temperature T_t and CO_2e concentration M_t for an imminent tipping scenario $T^c = 2°$ (black line, square markers), a remote tipping scenario $T^c = 3°$ (dark grey line, diamond markers), and no temperature feedback λ^{T^c} (light grey line, circle markers). The solid lines show attracting equilibria and the dotted lines show repelling equilibria.

relationship between temperature T_t and CO_2e concentration M_t is not affected before the temperature feedback comes into effect.[4] If CO_2e concentration M_t increases and temperature crosses the critical threshold T^c, the old stable and low temperature regime is no longer feasible and only a high stable temperature regime remains. Then, any increase of CO_2e concentration pushes the system past a tipping point and temperature rapidly converges to the high temperature equilibrium. Crucially, to revert the system back to the lower temperature regime, it is not sufficient to remove just the carbon that caused the system to tip, but to remove all carbon until the only stable equilibrium is the low temperature. In Figure 3, this would amount to bringing carbon back to $M_t < 550$, where the high temperature regime (solid black line) vanishes.

We can now ask what path of temperature T_t we would observe in these three scenarios if no abatement is implemented and CO_2e concentration M_t grows at the no-policy rate γ_t^{np}, such that $M_t = M_t^{\mathrm{np}}$. This is illustrated in Figure 4, which shows the median path over 60 years (thick lines with markers) of temperatures T_t and CO_2e concentration M_t under no-policy for the model without feedback (light grey line, circle markers), a remote tipping

[4]Technically, the stable manifold of the model without feedback is a contact element of the stable manifold in the case with feedback.

scenario $T^c = 3°$ (dark grey line, diamond markers), and an imminent tipping scenario $T^c = 2°$ (black line, square markers). Each marker is the outcome at the beginning of each decade. The simulations are overlaid onto the equilibria from Figure 3. In the case of $T^c = 2°$, the

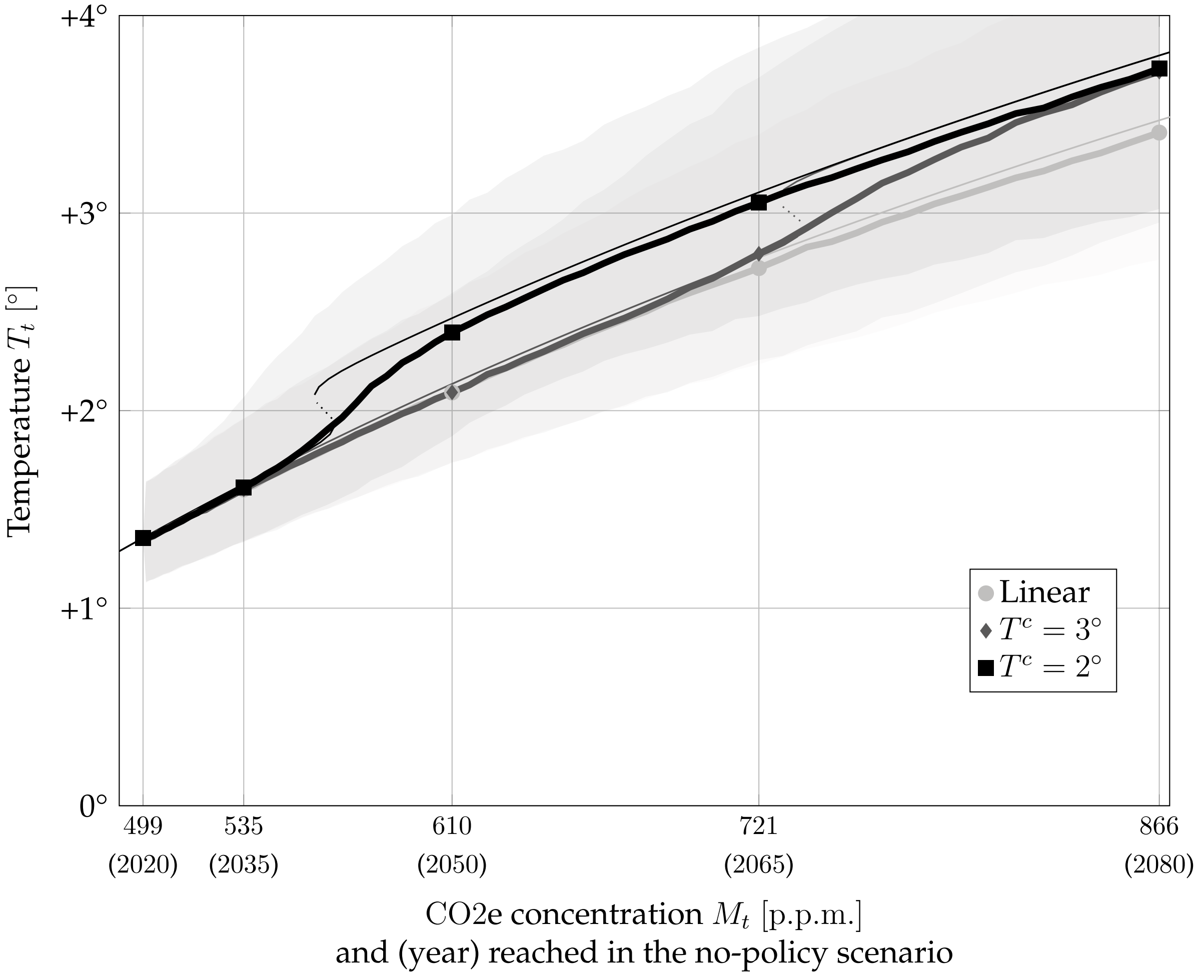

Figure 4: Median temperature T_t and CO_2e concentration M_t at time t of 100,000 simulations in case of no feedback (light grey line, circle markers), a remote tipping scenario $T^c = 3°$ (dark grey line, diamond markers), and an imminent tipping scenario $T^c = 2°$ (black line, square markers). The paths are simulated under the temperature dynamics (10) and the no-policy CO_2e concentration (1). The thinner lines represent the equilibrium relationships as in Figure 3, with solid branches attracting and dotted branches repelling.

temperature deviates from the other scenarios by 2030. In case of $T^c = 3°$, temperature growth cannot be distinguished from a situation without temperature feedback and the two diverge only starting in 2050. In both cases, once the tipping point is crossed and the only regime is the high temperature one, temperature grows quickly to the new equilibrium. After that, the linear relationship between temperature and (log) CO_2e concentration is restored.

2.2 Economy

This section introduces the economy. The model assumes an AK economy, following Pindyck and Wang (2013), and Hambel, Kraft and Schwartz (2021).

2.2.1 Climate Damages and Abatement Costs

Output Y_t, in trillion of US \$ per year, is the product of the capital stock K_t and its productivity A_t, that is,

$$Y_t = A_t K_t. \tag{12}$$

Climate change affects the economy by changing productivity growth $\frac{\mathrm{d}A_t}{\mathrm{d}t}$ via a damage function $d(T_t)$, which is otherwise assumed to grow at a constant exogenous rate ϱ, that is,

$$\frac{\mathrm{d}A_t}{\mathrm{d}t} = \varrho - d(T_t). \tag{13}$$

This is in line with recent empirical evidence on the channels by which warming affects the economy (Burke, Hsiang and Miguel, 2015; Dell, Jones and Olken, 2012; Kalkuhl and Wenz, 2020). Damages to productivity growth have been shown in the agricultural (Dietz and Lanz, 2019) and manufacturing sectors (Dell, Jones and Olken, 2009, 2012). Further discussion on this assumption is provided in B.2.3. I employ the quadratic global damage function estimated by Kalkuhl and Wenz (2020, Table 4, Section 6.2) and, hence, assume the damages to be quadratic in temperature T_t increases with respect to preindustrial level, that is,

$$d(T_t) := \xi T_t^2. \tag{14}$$

The estimated damage parameter is $\xi = 0.0709\%$. I choose this particular estimation as, conditional on the damage function (14) assumed here, the empirical strategy in Kalkuhl and Wenz (2020) relies on more recent and fine-grained data than Burke, Hsiang and Miguel (2015). Nevertheless, this is not the authors' preferred specification, as they find additional channels of temperature growth and precipitation growth and level on output growth. The damage function employed here, compared to other popular estimates in the literature, is displayed in Figure 5. I refer the reader to B.2.3 for a comparison of the damage function employed here with popular level damage functions and to Hambel, Kraft and Schwartz (2021) for a discussion on the broader impacts of level or growth damages in integrated assessment models.

Output Y_t can be used for investment in capital I_t, abatement expenditures B_t, or consumption C_t. This entails the budget constraint

$$Y_t = I_t + B_t + C_t. \tag{15}$$

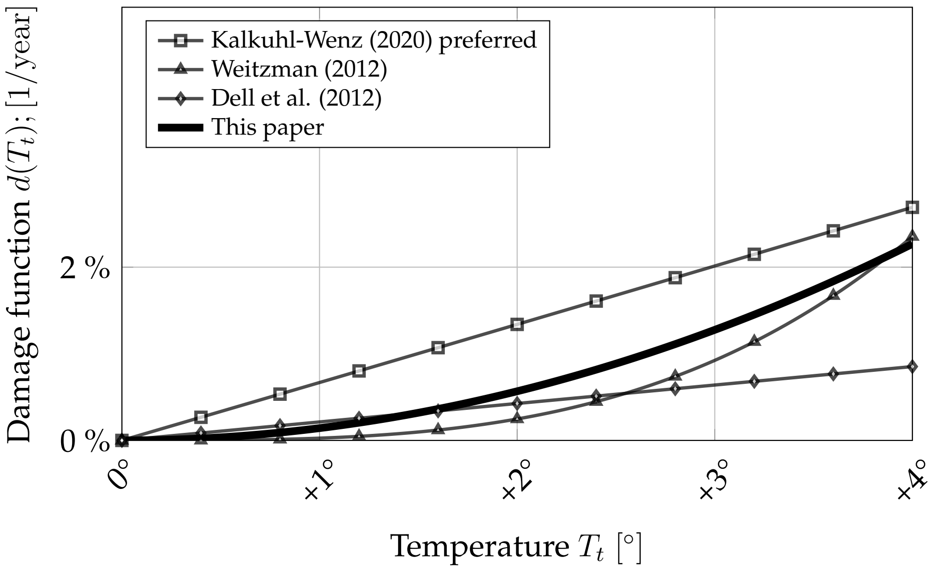

Figure 5: Damage function calibration from this paper compared to Kalkuhl and Wenz (2020)'s preferred calibration, Weitzman (2014), and Dell, Jones and Olken (2012).

Capital K_t depreciates at a rate δ_k but can be substituted by capital investments I_t, which are subject, along with abatement expenditure B_t, to quadratic adjustment costs

$$\frac{\kappa}{2}\left(\frac{I_t + B_t}{K_t}\right)^2 K_t. \tag{16}$$

Combining these, the endogenous growth of capital satisfies

$$\frac{\mathrm{d}K_t}{K_t} = \left(\frac{I_t}{K_t} - \delta_k - \frac{\kappa}{2}\left(\frac{I_t + B_t}{K_t}\right)^2\right)\mathrm{d}t + \sigma_k \mathrm{d}W_{k,t}, \tag{17}$$

where $W_{k,t}$ is a standard Brownian motion.

In the following, I link the abatement costs B_t with the abatement rate α_t, introduced in the previous section. Define $\beta_t \coloneqq B_t/Y_t$ to be the fraction of output devoted to abatement. I assume this to be a power function of the fraction of abated emissions $\varepsilon(\alpha_t)$ (4), namely,

$$\beta_t(\varepsilon(\alpha_t)) = \omega_t\, \varepsilon(\alpha_t)^b. \tag{18}$$

Under this assumption, no abatement is free, as $\beta_t(0) = 0$. At a fixed time t, a higher abatement rate α_t and hence a higher emission reduction $\varepsilon(\alpha_t)$ vis-à-vis the no-policy scenario, becomes increasingly costly at rate

$$\beta_t'(\varepsilon(\alpha_t)) = b\, \omega_t\, \varepsilon(\alpha_t)^{b-1}. \tag{19}$$

It is common in the literature to assume the marginal abatement costs to be proportional to output and a power function in the abatement efforts (Baker, Clarke and Shittu, 2008; Dietz and Venmans, 2019; Hambel, Kraft and Schwartz, 2021; Nordhaus, 1992, 2017). As noted by Dietz and Venmans (2019, p.112-113), the proportionality with output arises because higher output growth increases energy demand. This must be satisfied with low-carbon energy technology which displays decreasing marginal productivity.

As time progresses, so does abatement technology and a given abatement objective becomes cheaper, as a fraction of output. This is modelled by letting the exogenous technological parameter ω_t decrease exponentially over time

$$\omega_t = \omega_0 e^{-\rho_\omega t}, \tag{20}$$

at a rate ρ_ω. The parameters ρ_ω and ω_0 are calibrated to match the implied marginal abatement curves in the Sixth Assessment Report of the IPCC (2023*b*, Figure 3.33). More details on the calibration procedure can be found in B.2.2. The resulting abatement curves are displayed in Figure 6. The shaded area represents the costs associated with abating more than 100% of the

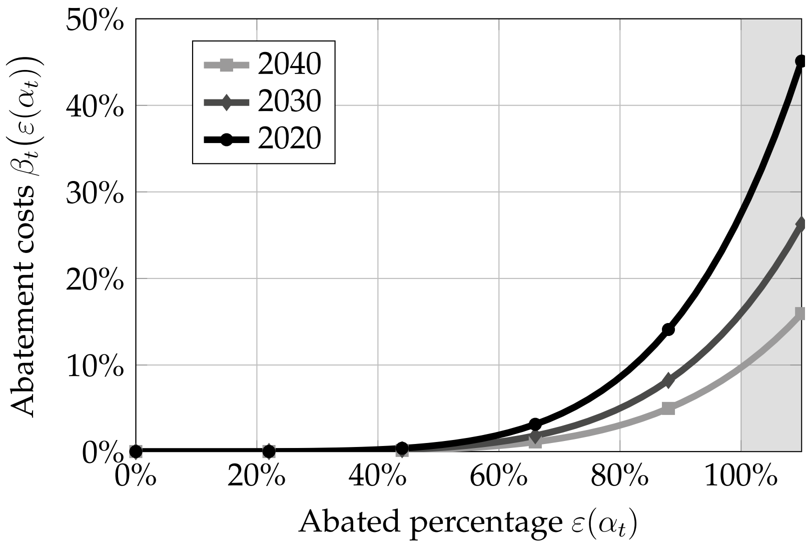

Figure 6: Calibrated abatement curves $\beta_t'(\varepsilon)$ for different times t.

no-policy emissions. In this case, CO_2e is sequestered from the atmosphere.

Finally, let

$$\chi_t := \frac{C_t}{Y_t} \tag{21}$$

be the fraction of output devoted to consumption. Using the budget constraint (15) and the two controlled rates, the abatement efforts α_t and the consumption rate χ_t, the growth rate of output can be rewritten, in terms of growth rates, as

$$\frac{\mathrm{d}Y_t}{Y_t} = \Big(\underbrace{\varrho + \phi_t(\chi_t)}_{\text{growth}} - \overbrace{A_t \beta_t(\varepsilon(\alpha_t))}^{\text{abatement}} - \underbrace{d(T_t)}_{\text{climate}} \Big) \mathrm{d}t + \sigma_k \mathrm{d}W_{k,t}, \tag{22}$$

where

$$\phi_t(\chi_t) \coloneqq A_t(1 - \chi_t) - \frac{\kappa}{2} A_t^2 (1 - \chi_t)^2 - \delta_k \tag{23}$$

is an endogenous growth component, $\beta_t(\varepsilon(\alpha_t))$ is the fraction of output allocated to abatement for an abatement rate α_t, and $d(T_t)$ are the damages from climate change.

This formulation models the trade-off between climate abatement and economic growth. On the one hand, devoting fewer resources to abatement (lower α_t and, hence, $\beta_t(\varepsilon(\alpha_t))$) to pursue higher output growth, yields higher future temperature and can put the economy in a lower growth path altogether (higher $d(T_t)$). On the other hand, more ambitious abatement lowers future temperature and climate damages (lower $d(T_t)$), which boosts future economic growth, but costs economic growth today (higher α_t and, hence, $\beta_t(\varepsilon(\alpha_t))$).

3 Optimal Abatement Under Complete Information

Given the climate and economic dynamics described in the previous section, I introduce the objective and optimal policies of a social planner who knows the tipping point location. I refer to this case as a "complete information". Beyond the Knightian uncertainty in the tipping point, there is risk in the model, as temperature T_t, carbon concentration M_t, and output Y_t are subject to shocks. This risk is fully quantifiable and the social planner, under complete information, can maximise the average stream of utility. To account for relative risk aversion and intertemporal elasticity substitution the utility is recursively defined using the Epstein-Zin aggregator (Duffie and Epstein, 1992). Formally, at time t, given the state of temperature, $\mathrm{CO_2e}$ concentration, and output $X_t \coloneqq (T_t, M_t, Y_t)$, an abatement schedule α and a consumption schedule χ, societal utility is recursively defined as

$$W_t^{T^c}(X_t; \alpha_t, \chi_t) = \mathbb{E}_t \int_t^\infty f(\chi_s(X_s)Y_s, W_s(X_s; \alpha_s, \chi_s)) \mathrm{d}s, \tag{24}$$

subject to the state dynamics (3), (10), and (22) under critical threshold T^c, where f is the Epstein-Zin aggregator

$$f(c, w) \coloneqq \rho \frac{(1-\theta)\ w}{1-1/\psi} \left(\left(\frac{c}{((1-\theta)w)^{\frac{1}{1-\theta}}} \right)^{1-1/\psi} - 1 \right). \tag{25}$$

This aggregator plays a dual role. First, it disentangles the role of relative risk aversion θ, elasticity of intertemporal substitution ψ, and the discount rate ρ in determining optimal abatement paths. Second, it circumvents the known paradoxical result that abatement policies become less ambitious as society becomes more risk averse (Pindyck and Wang, 2013). Furthermore, I denote the maximal welfare by V, which satisfies

$$V_t^{T^c}(X_t) = \sup_{\chi, \alpha} \mathbb{E}_t \int_t^\infty f(\chi_s(X_s) Y_s, V_s(X_s)) \mathrm{d}s, \tag{26}$$

where χ and α are continuous policies over time and the state space.

Following the baseline formulation in Hambel, Kraft and Schwartz (2021), I set $\rho = 1.5\%$, $\theta = 10$ and $\psi = 1$. There is no consensus around the calibration of these parameters. I refer the reader to Pindyck and Wang (2013, Section 2), for a detailed discussion on the calibration of the preference parameters, and to Hambel, Kraft and Schwartz (2021) for their role in determining optimal abatement. Furthermore, the formulation assumes that preferences are stable over time, both in the parameters θ, ψ, and ρ and in their depending only on the consumption stream C_t. Future society might exhibit different preferences, for example, there might be an inherent utility from lower $\mathrm{CO_2e}$ concentration or temperature levels. For a discussion on incorporating this possibility in the analysis I refer the reader to Le Kama and Schubert (2004).

The social planner problem (26) is solved numerically. For more details, see C. The numerical solution relies on three steps. First, I derive the Hamilton-Jacobi-Bellman equation for the value function of the social planner V_t (26) and show that it can be factorised in an output dependent component Y_t and a cost function $F_t(T_t, M_t)$, which satisfies an auxiliary Hamilton-Jacobi-Bellman equation[5]. Second, in Section C.2, I derive an approximating Markov chain, that is, a discretisation of the state space T_t, M_t and a discrete Markov chain F_t^h over the discretisation, parametrised by a small step $h > 0$. I show that the recursive definition for F_t^h converges to the Hamilton-Jacobi-Bellman equation as $h \to 0$. This step extends the method

[5]This factorisation comes from the homogeneity of the Epstein-Zin aggregator f in Y, and it is commonly employed in the literature (Hambel, Kraft and Schwartz, 2021). I thank Christoph Hambel for pointing this out to me.

developed by Kushner and Dupuis (2001) to a problem with recursive utility. Third, to compute the discretised cost function F^h_t, I assume that at $t = \tau \equiv 500$, full abatement is achieved $\alpha_\tau = \gamma^{np}_\tau$ and compute the discretised terminal cost function F^h_τ by value function iteration. Finally, by backward induction, I compute the discretised cost function F^h_t and the optimal policies α_t, χ_t using their first-order conditions. The model was solved on the Snellius HPC cluster using a parallelisation technique based on Bierkens, Fearnhead and Roberts (2019). The method developed by Kushner and Dupuis (2001) and adapted here, discretises directly the recursive definition of the value function (26) rather than the Hamilton-Jacobi-Bellman equation, as it is commonly done in finite difference schemes. Furthermore, heuristically, the method performs well in the presence of nonlinearities, such as the one displayed by the temperature dynamics (10), as it produces a discrete Markov chain, analogous to the discrete time one in Cai and Lontzek (2019), with time steps proportional to the drift's magnitude in the state dynamics, such that around the tipping point the discretisation is finer in time and, hence, more precise.

Under the preference parameter choices described above, the solution, given a critical threshold T^c, yields a closed-loop abatement policy function φ^{T^c} such that optimal abatement is given by

$$\alpha_t = \varphi^{T^c}(t, T_t, M_t). \tag{27}$$

For illustration purposes, in the following I show optimal policies and their consequences in two climate scenarios: a "tipping" climate with a critical threshold of $T^c = 2.5°$ and a "linear" climate without a tipping point. The optimal abatement policies imply an optimal path of abatement ε_t and, as a consequence, of $\mathrm{CO_2e}$ concentration M_t and temperature T_t. As the processes of $\mathrm{CO_2e}$ concentration (3) and temperature (10) are stochastic, so is the optimal path of abatement. In the following simulation, the median and 95% quantiles of the variables are computed by first simulating 100,000 paths under $\mathrm{CO_2e}$ concentration M_t (3) and temperature T_t (10) dynamics with optimal abatement policies $\alpha_t = \varphi^{T^c}(t, T_t, M_t)$. Then at each moment t, the relevant quantiles of each variable are computed, which yields a quantile path in t for each variable. Figure 7 shows the median path of the optimal fraction of abated emissions for the linear (grey) and tipping (black) climate. In case of a linear climate, optimal abatement in 2020 is around 37% of the emissions in the no-policy benchmark, which corresponds to an emission level of approximately $36.77\mathrm{Gt\,CO_2e/year}$[6]. Over time, abatement

[6]Observed emissions in 2020 were approximately $54.5\mathrm{Gt\,CO_2e/year}$ (, 2023) and emissions under the SSP5-8.5 would have been approximately $58.37\mathrm{Gt\,CO_2e/year}$, indicating that the current policy trajectory falls in between no-policy and an optimal policy in a linear climate.

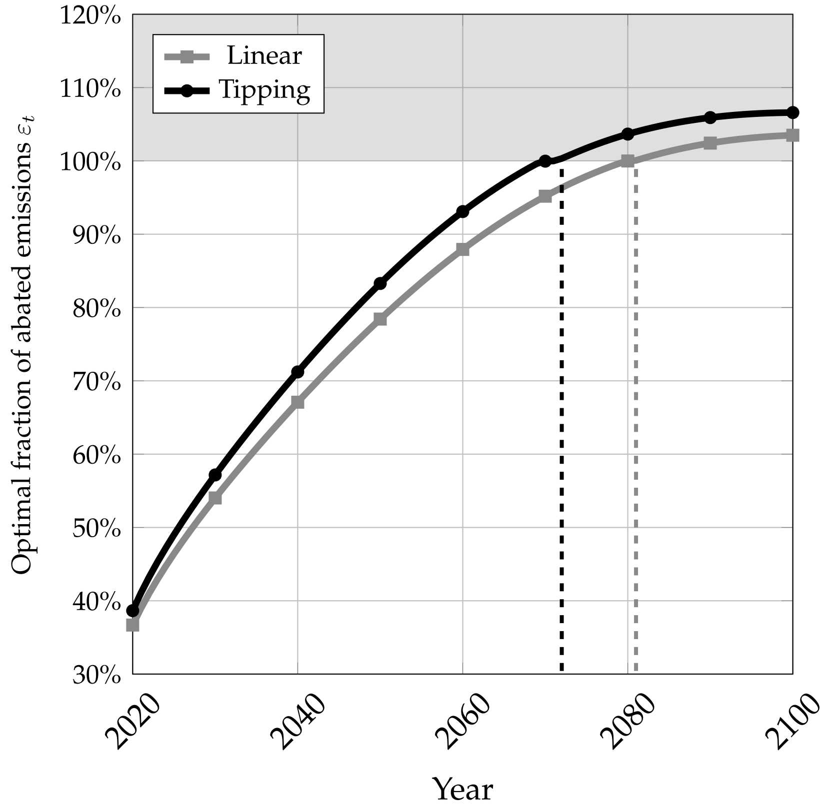

Figure 7: Median optimal abated fraction of emissions ε_t at time t of 100,000 simulations in a linear (grey) and a tipping climate (black).

increases until full abatement is reached in 2081. Thereafter, abatement is further scaled up above 100%. In case of a tipping climate, optimal abatement rises more rapidly than in a linear climate, reaching 80% of abatement by 2050 and 100% of abatement by 2072, 9 years earlier than with a linear climate. Furthermore, negative abatement efforts are stronger and more prolonged. Figure 8 shows the resulting optimal path of $\mathrm{CO_2e}$ variables in a linear (left panel) and a tipping (right panel) climate. The linear abatement policies cause $\mathrm{CO_2e}$ concentration M_t to peak in the 2060s just below 560 ppm before starting to decrease. As a consequence, temperature T_t in the median scenario are kept below 2° above pre-industrial levels. On the other hand, in a tipping climate, $\mathrm{CO_2e}$ concentration M_t peaks around 2070 below 540 $\mathrm{p.p.m.}$ and temperature T_t in the median scenario is kept well below 2° above pre-industrial levels, before converging back to around 1.5°. In the latter case, it is optimal for the social planner to quickly "slam the breaks" to avoid crossing a critical threshold and triggering a tipping point. Despite the tipping point not being triggered under optimal policies it still influences temperature T_t, as positive temperature shocks are more persistent. This can be seen in Figure 8 as the distribution of temperature is skewed towards higher temperatures whenever the climate is close to the tipping point, that is, between 2050 and 2080. Furthermore, once the tipping point is avoided, the planner keeps scaling up abatement efforts and

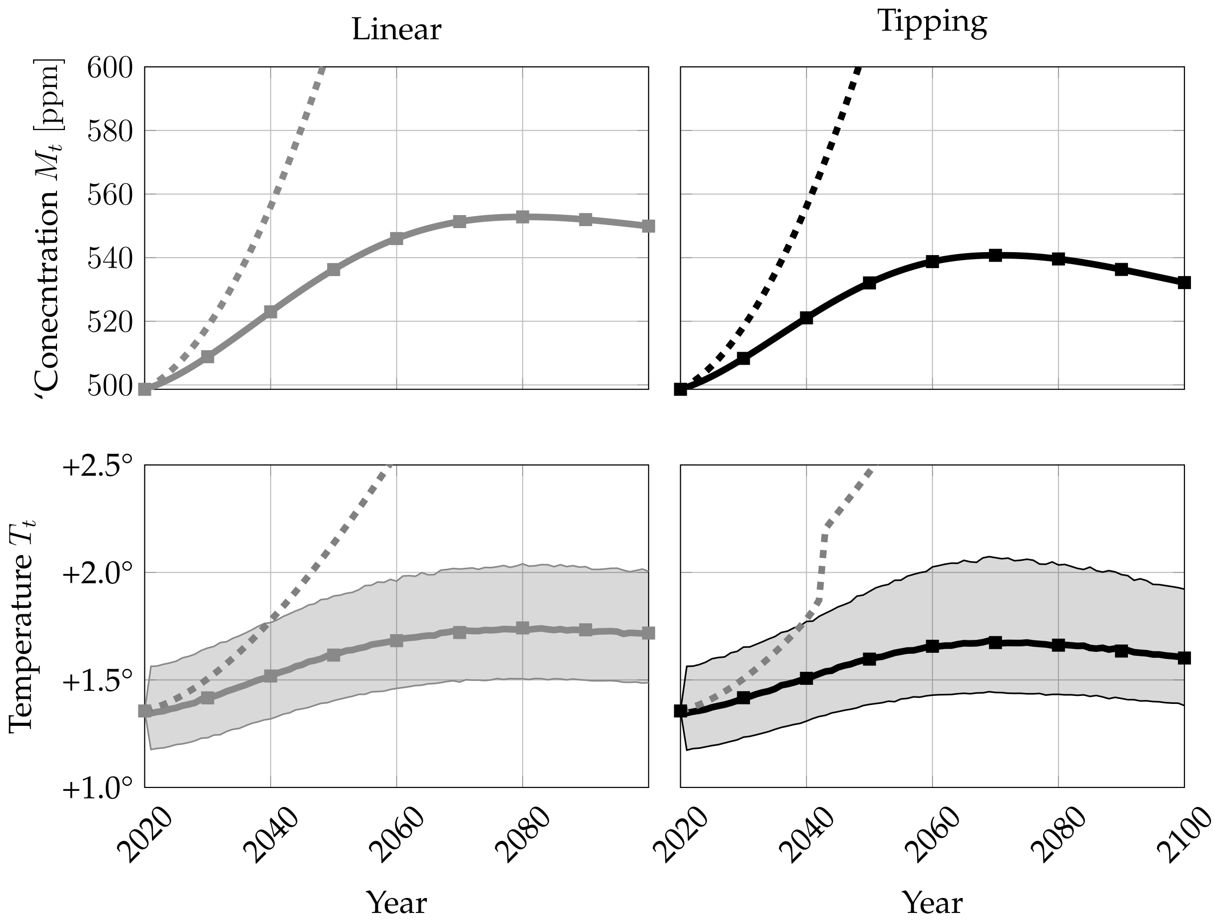

Figure 8: Median (solid) and 95 % simulation intervals (shaded area) of $\mathrm{CO_2e}$ concentration M_t and temperature T_t at time t of 100,000 simulations in a linear (left column) and a tipping (right column) climate. The paths are simulated under the temperature (10), $\mathrm{CO_2e}$ concentration (3), and output (22) dynamics with optimal abatement φ^{T^c}. The dashed line represents the median path in the no-policy scenario.

stabilises temperatures around 1.5° to move away from the critical temperature threshold, hence reducing the probability of a positive temperature shock triggering a tipping point.

To tie these trajectory outcomes back to the underlying decision rule, Figure 9 shows the fraction of abated emission ε_t resulting from the implementation of the optimal policy $\varphi^{2.5°}$ in the tipping climate (black curve) and the optimal policy $\overline{\varphi}$ in the linear climate (grey curve), along the no-policy path of CO_2e concentration. In Figure 9, the optimal fraction of abated

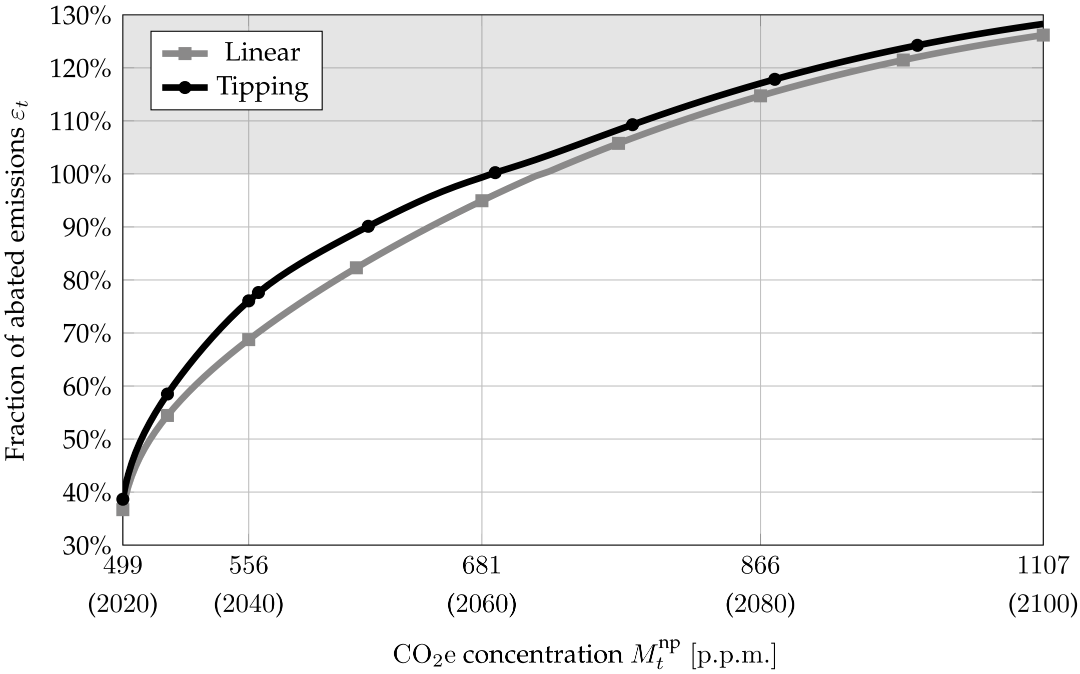

Figure 9: Median optimal abated fraction of emissions ε_t along the no-policy path of CO_2e concentration M_t^{np} of 100,000 simulations using the optimal policy $\overline{\varphi}$ in a linear climate (grey) and $\varphi^{2.5°}$ in a tipping climate (black).The shaded area represents negative emissions.

emission ε_t is displayed as a function of the carbon concentration along the no-policy np path, that is, for a given level of CO_2e concentration M_t^{np}, I compute the associated equilibrium temperature T_t^{np} and the time t at which it would be reached in the np scenario, hence, in absence of prior abatement efforts. Then the optimal fraction of abated emission $\varepsilon_t = \varepsilon(\varphi^{T^c}(t, T_t^{np}, M_t^{np}))$ is computed. In the presence of a tipping climate (black curve) the optimal fraction of abated emission ε_t responds more strongly to an increase CO_2e concentration, compared to a linear climate (grey curve). Once the tipping point is crossed abatement efforts are scaled back (post-tipping) as the climate is in a regime of high temperature. This result stems from the dual role of abatement: reducing first order climate damages and preventing the climate from tipping. Around 2040, once the tipping point is crossed, the latter motive vanishes, and the response of optimal abatement to an increase in CO_2e concentration decreases closer to the optimal policy in a linear climate (grey curve). This policy-function shift helps explain why the trajectories in Figure 8 diverge most strongly before the tipping region is reached. This result echoes that of Clarke and Reed (1994), in which the authors shown that abatement efforts are scaled back once catastrophic risk becomes unavoidable.

The above simulations provide a qualitative description of the two scenarios, but to directly compare them it is necessary to take into account output growth and societal preferences. To do so, I compute the social cost of carbon (SCC) for the linear climate $\overline{\mathrm{SCC}}$ and for the tipping climate SCC^{T^c} at different critical thresholds T^c. The SCC, measured in US\$ per $\mathrm{t\,CO_2e}$ is the marginal rate of substitution between carbon equivalent emissions and output. This can also be interpreted as an optimal carbon tax and hence used to compare different climate regimes. In the context of our model, it can be computed as

$$\mathrm{SCC}_t = -\frac{\partial V_t / \partial E_t}{\partial V_t / \partial Y_t}. \tag{28}$$

Figure 10 shows the path of the SCC_t in a linear and a tipping climate. Under both climates,

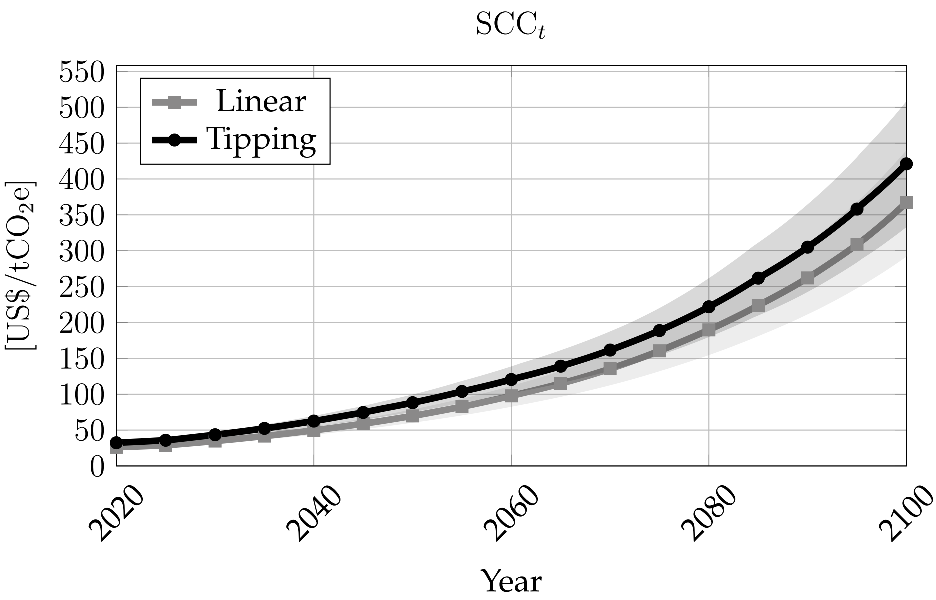

Figure 10: Median (solid) and 95 % simulation intervals (shaded area) path of social cost of carbon $\mathrm{SCC}^{T^c=2.5°}$ in a tipping and $\overline{\mathrm{SCC}}$ in a linear climate.

the social cost of carbon grows rapidly from its initial value of around 25 $\mathrm{US\$/t\,CO_2e}$, driven by the increased marginal damages of $\mathrm{CO_2e}$ emissions as temperature increases and output growth. The social cost of carbon in a tipping climate is higher throughout, reflecting the possibility of a tipping point being triggered. This difference grows rapidly between 2020 and 2080, when, in case of a tipping climate, the social planner quickly abates emissions to avoid triggering a tipping point.

The social cost of a carbon at the start of the period, that is $t = 2020$, is a measure of the cost of tipping at different critical thresholds T^c. Figure 11 shows $\mathrm{SCC}^{T^c}_{t=2020}$ as a function of different critical thresholds. The dashed line represents the social cost of carbon in a linear climate. First, the social cost of carbon $\mathrm{SCC}_{t=2020}^{T^c}$ is strictly decreasing in the critical threshold

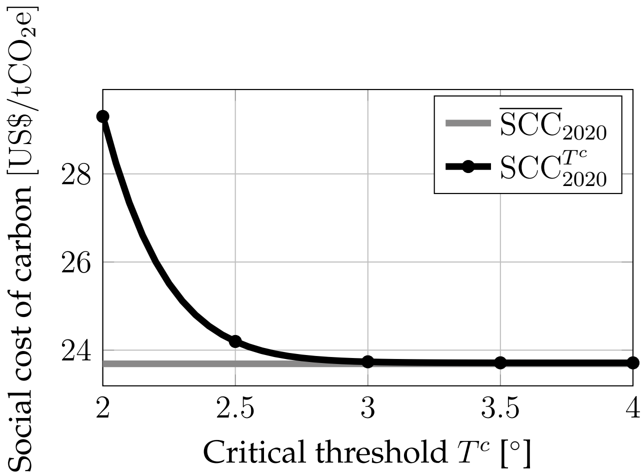

Figure 11: Social cost of carbon $\mathrm{SCC}_t^{T^c}$ at the start of the period $t = 2020$, for different critical thresholds T^c. The dashed line represents the social cost of carbon in a linear climate.

T^c, as a higher critical threshold gives the social planner more time to abate. This effect is particularly pronounced for critical thresholds $T^c < 2.5°$ as, for higher critical thresholds, direct climate damages are sufficient incentives to optimally fully abate $\mathrm{CO_2e}$ emissions at temperature levels T_t lower than the critical threshold. Nevertheless, the presence of a tipping point, even if only triggered for large temperature deviations T_t, yields a larger $\mathrm{SCC}_{t=2020}^{T^c}$ compared to a linear climate, because there is always a probability that a large temperature shock triggers a tipping point. As $T^c \to \infty$, this probability becomes vanishingly small and the social cost of carbon in a tipping climate converges to the linear one, that is, $\mathrm{SCC}_t^{T^c} \to \overline{\mathrm{SCC}}_t$. Second, the presence of a tipping point can significantly worsen the cost of climate change, as reflected in the $\mathrm{SCC}_{t=2020}^{T^c}$ at the lowest threshold $T^c = 2°$ being approximately 26% larger than in the linear climate. Yet, the social cost of carbon is highly sensitive to the location of the critical threshold: a shift of $0.5°$ in T^c from $2°$ to $2.5°$ causes the social cost of carbon at $t = 2020$ to drop from 26% above the linear climate baseline to only around 2% above it. Hence, knowing the location of the critical threshold T^c is a major factor in optimally schedule abatement policies and small mistakes can lead to large consequences. Unfortunately, in the climate literature there is no consensus on whether we can predict such critical thresholds (Ben-Yami et al., 2024; Ditlevsen and Ditlevsen, 2023; Smolders, van Westen and Dijkstra, 2024), especially as the climate warms quickly due to anthropogenic greenhouse gasses emissions (Rietkerk et al., 2025).

4 Robust Policy and the Uncertainty Premium

In this section, I focus on the uncertainty in the tipping element's critical threshold T^c. Given the aforementioned lack of consensus in the climate literature, I assume the planner cannot form a probabilistic prior over T^c. Instead, the planner treats the uncertainty around T^c as Knightian and chooses their abatement policy to minimize the maximum regret (min-max regret) as proposed by Savage (1951). When evaluating a policy, the planner computes how much higher welfare would have been in each possible state of the world and attempts to minimise the welfare gap, that is, the regret, in the worst state of the world (Heal and Millner, 2014). Crucially, the policy does not require the planner to assign probabilities to each state of the world. Such approach has been recently employed in climate economics in dealing with Knightian uncertainty around climate models (DeCanio, Manski and Sanstad, 2022; Rezai and Van Der Ploeg, 2017). The following formalises this idea in the context of the model, introduces the regret associated with a policy φ, and defines the robust policy of the planner φ^r.

Given an abatement policy φ and a critical threshold T^c, the regret at time t and state X_t is given by

$$\mathcal{R}(\varphi, T^c; t, X_t) \coloneqq V_t^{T^c}(X_t) - W_t^{T^c}(X_t; \varphi), \tag{29}$$

where V is the societal optimal welfare (26) and W is the societal welfare (24) conditional on a policy φ. As above, we assume that the policy φ is a continuous function of time t and the state X_t. The dependence of W on the consumption path χ is omitted as, under the baseline specification, the optimal consumption path depends only on time t and not on the state X_t (C.1.2). Notice that regret $\mathcal{R}$ is non-negative and is zero only at the optimal policy φ^{T^c}.

The robust policy φ^r is then defined as follows. First, I discretise the space of possible critical threshold, that is, I assume that

$$T^c \in \mathcal{T} \coloneqq \{\underline{T}^c, \underline{T}^c + \Delta T^c, \underline{T}^c + 2\Delta T^c, \dots, \overline{T}^c, \infty\} \tag{30}$$

with some discretisation step ΔT^c and where $T^c = \infty$ denotes the linear model, without tipping elements. I choose $\underline{T}^c = 2°$, $\overline{T}^c = 5°$, and $\Delta T^c = 0.025°$. Then, I assume that, in choosing the robust policy φ^r, the social planner is constrained to choose a convex combination of optimal policies

$$\Phi := \left\{ \varphi \;\middle|\; \varphi_t(X_t) = \sum_{T_i^c \in \mathcal{T}} w^{T_i^c} \varphi_t^{T_i^c}(X_t) \text{ where } w^{T_i^c} \geq 0 \text{ and } \sum_{T_i^c \in \mathcal{T}} w^{T_i^c} = 1 \right\}. \tag{31}$$

This assumption, on the one hand, makes the problem computationally feasible and, on the other hand, ensures that the robust policy is a principled hedge across threshold scenarios. Using this, I define the robust policy as

$$\varphi^r := \arg\min_{\varphi \in \Phi} \left\{ \max_{T^c \in \mathcal{T}} \mathcal{R}(\varphi, T^c; 0, X_0) \right\}. \tag{32}$$

The robust policy is then a linear combination of optimal policies

$$\varphi_t^r(X_t) = \sum_{T_i^c \in \mathcal{T}} w^{T_i^c} \varphi_t^{T_i^c}(X_t). \tag{33}$$

Importantly, the weights $w^{T_i^c}$ are independent of time t and state X_t, as they are chosen based on the initial regret $\mathcal{R}(\varphi, T^c; 0, X_0)$, but the robust policy φ^r depends on time t and state X_t via the optimal policies $\varphi^{T_i^c}$. This is consistent with the Knightian formulation of the problem, as the only learning possible within the model is about the location of the critical threshold within $\mathcal{T}$, namely through ruling out thresholds below the realised temperature path, rather than through probabilistic updating over T^c. Furthermore, as illustrated in Figure 13 below, in the median scenario the robust policy stabilises temperature before the assumed lower bound on the critical threshold $\underline{T}^c$, hence the set of possible thresholds $\mathcal{T}$ in equilibrium does not shrink over time. Along this equilibrium path, there is therefore no additional learning that would justify time-varying weights. One limitation of this assumption is that it does rule out the possibility that the planner might purposefully induce higher temperatures in order to shrink $\mathcal{T}$. Nevertheless, I consider this possibility unrealistic. More details on the computation of the robust policy φ^r are provided in D.1.

Table 1 shows the 5 largest weights $w^{T_i^c}$, which account for 90.4% of the total weight, and the corresponding optimal policy's critical threshold T_i^c. The fact that most weights are at-

T^c	$2°$	$4°$	$2.15°$	$3.3°$	$3.65°$
$w^{T_i^c}$	34%	31.3%	11.2%	7.7%	6.2%

Table 1: Weights of the robust policy across critical thresholds, sorted by weight.

tached to this subset of scenarios suggests that these scenarios capture the salient features of the space of optimal policies. The two largest weights are placed on two extreme scenarios: a critical threshold $T^c = 2°$ and a remote one $T^c = 4°$. The former ensures that the policy φ^r is robust to the regret from possibly under-abating in an environment in which a critical threshold is imminent, thereby triggering a tipping point. The latter ensures that the policy φ^r is robust to the regret from over-abating and quickly reducing emissions in a scenario in which the critical threshold is high enough to not be binding. The remaining weight, associated with $T^c \in \{2.15°, 3.3°, 3.65°\}$, spans intermediate scenarios. I now illustrate the robust policy φ^r and how it affects the path of $\mathrm{CO_2e}$ concentration M_t and temperature T_t. In the following, I denote the evolution under the robust policy by “Robust” and compare it to the “Linear” and “Tipping” scenarios introduced in the previous section. Figure 12 displays the optimal fraction of abated emissions ε_t over time, similarly to Figure 7. The robust pol-

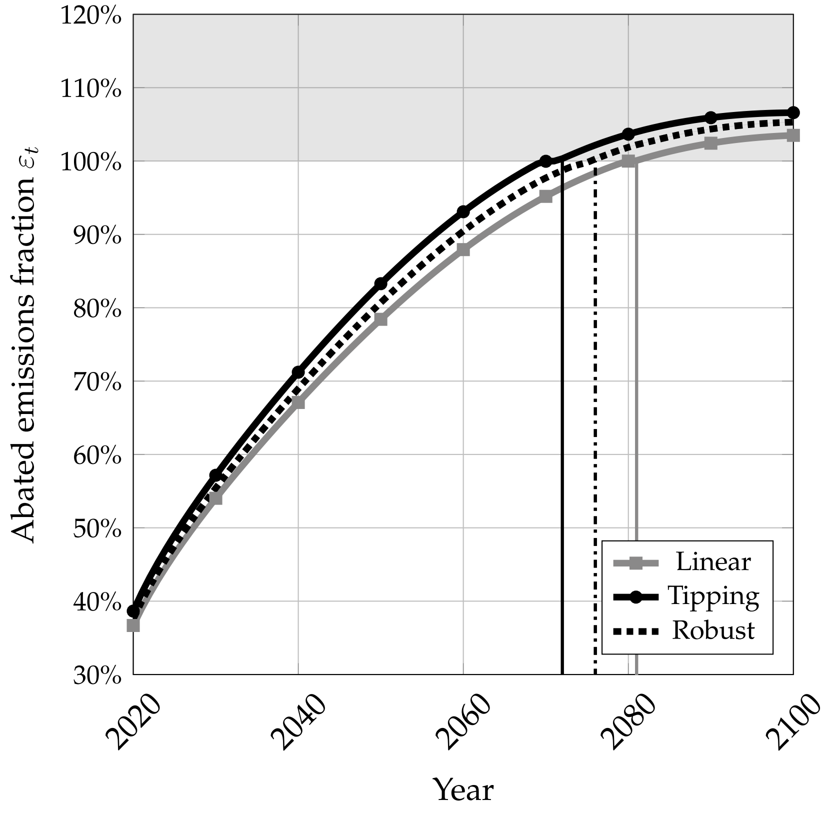

Figure 12: Median optimal abated fraction of emissions ε_t over time t of 100,000 simulations using the optimal policy in a linear climate (grey), in a tipping climate (black), and the robust policy (dashed).

icy φ^r prescribes abatement that lies strictly between the linear and tipping optimal policies throughout, reflecting its construction as a convex combination of the latter. Under robust abatement, the climate is stabilised in 2077, 4 years before the optimal linear policy and 5 years after the optimal tipping policy. The earlier stabilisation relative to the linear scenario is driven by the large weight placed on scenarios with low critical thresholds (Table 1). Anticipating the possibility of an imminent tipping point, the planner commits to a more aggressive decarbonisation path even when the true threshold may remain non-binding.

Figure 13 traces the consequences of the three abatement policies for the resulting CO_2e concentration M_t and temperature T_t paths, similarly to Figure 8. Under both climate scenarios, the robust policy φ^r generates CO_2e concentration and temperature paths that lie between those induced by the linear and tipping optimal policies, consistent with its intermediate abatement intensity. In the linear scenario (left column), the higher abatement of φ^r relative to $\overline{\varphi}$ produces a lower peak CO_2e concentration of approximately 550p.p.m. around 2070 and stabilises temperatures well below 2°. In the tipping scenario (right column), the robust policy closely tracks the tipping optimal φ^{T^c} for the first couple decades before stabilising the climate at a higher temperature level. At this temperature level, the 95% simulation intervals are skewed towards higher temperatures, reflecting the additional persistence induced by the closer proximity to the critical threshold T^c.

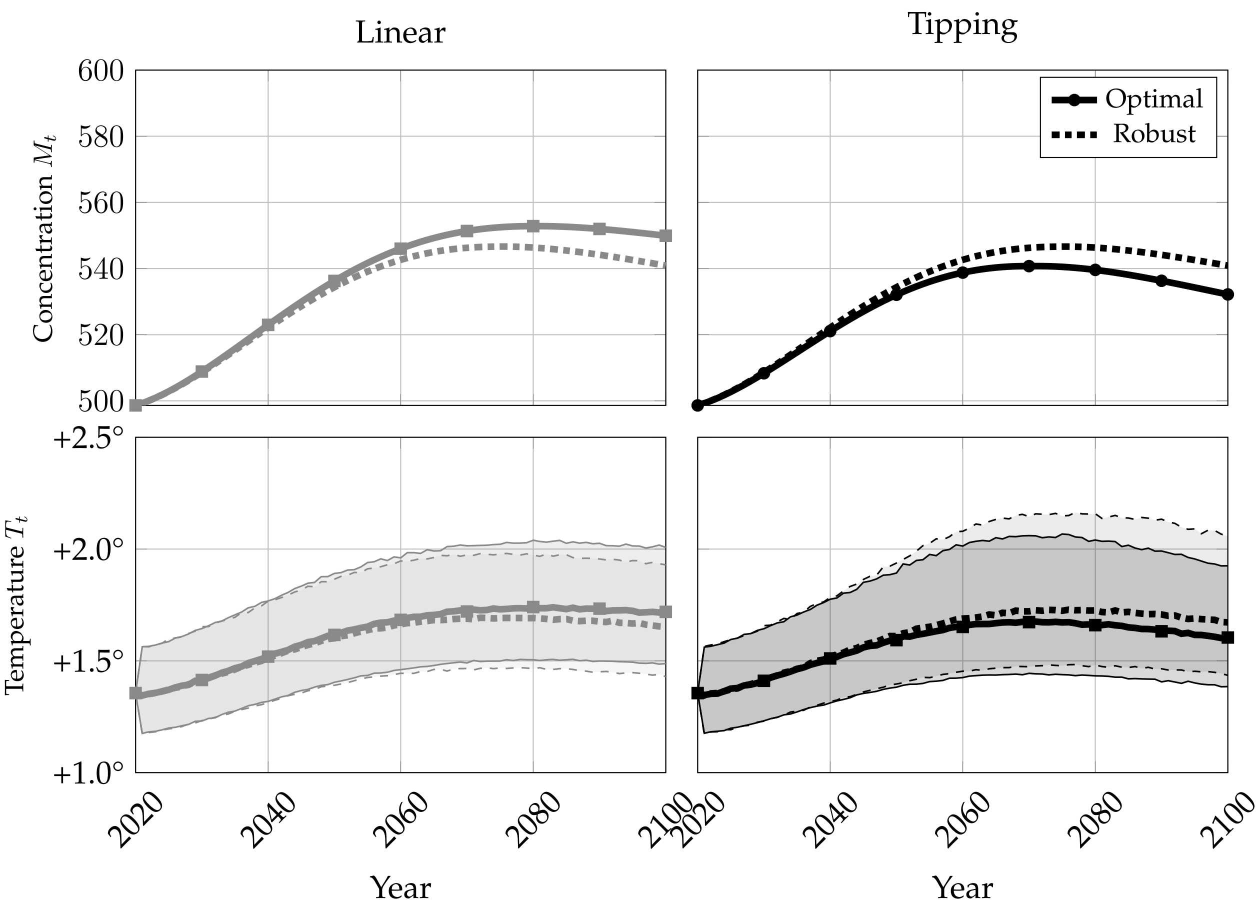

Figure 13: Median and 95% simulation intervals of CO_2e concentration M_t (ppm, top row) and temperature T_t (bottom row) over time t of 100,000 simulations in a linear climate (left column) and a tipping climate (right column). Solid lines and darker shaded bands show the optimal policy φ^{T^c}; dash-dotted lines and lighter shaded bands show the robust policy φ^r.

Figure 14 overlays the robust policy φ^r (dashed) over the no-policy np path, to the optimal policies in the linear $\overline{\varphi}$ (grey) and tipping φ^{T^c} case, as displayed in Figure 9. First, the robust policy φ^r entails strictly larger emission abatement than the optimal policy in the linear case

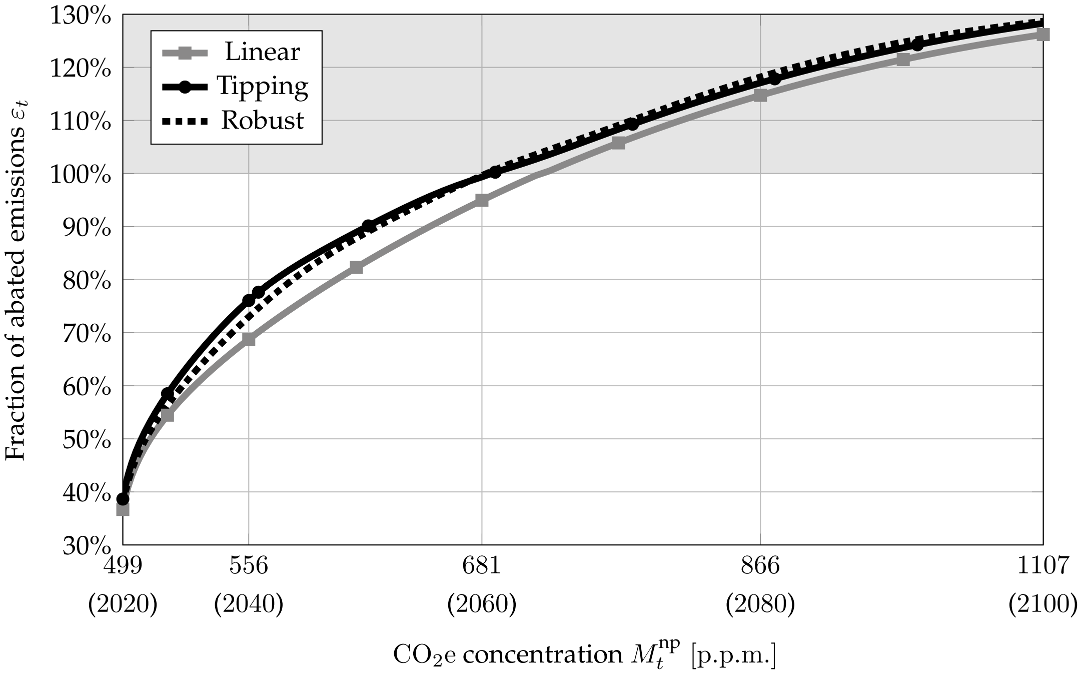

Figure 14: Median optimal abated fraction of emissions ε_t along the no-policy path of CO_2e concentration M_t^{np} of 100,000 simulations using the optimal policy $\bar{\varphi}$ in a linear climate (grey), φ^{T^c} in a tipping climate (black), and the robust policy φ^r (dashed). The marker represents the CO_2e concentration at which the no-policy np temperature T is at the critical threshold $T^c = 2.5°$ of the tipping climate. The shaded area represents negative emissions.

$\overline{\varphi}$. The gap between the two is driven by the precautionary motive of the planner who anticipates the presence of a critical threshold that it cannot foresee. Furthermore, the robust policy φ^r prescribes less abatement around the critical threshold than the optimal policy φ^{T^c}, and more abatement after the critical threshold is crossed, as it anticipates the possibility of critical thresholds at higher temperature levels.

Having computed the robust policy φ^r and the associated min-max regret

$$\mathcal{R}_t^r \coloneqq \min_{\varphi \in \Phi} \max_{T^c \in \mathcal{T}} \mathcal{R}(\varphi, T^c; t, X_t), \tag{34}$$

I now define the tipping point uncertainty premium on the social cost of carbon. Let $T_t^{c,r}$ denote the critical threshold with maximal regret under the robust policy, that is,

$$T_t^{c,r} \coloneqq \arg\max_{T^c \in \mathcal{T}} \mathcal{R}(\varphi^r, T^c; t, X_t), \tag{35}$$

so that $\mathcal{R}_t^r = V_t^{T_t^{c,r}}(X_t) - W_t^r(X_t)$. The uncertainty premium $\mathcal{P}_t$ is then defined as the relative difference between the social cost of carbon under the robust policy and the social cost of carbon under the complete information optimal policy at the threshold $T_t^{c,r}$, that is,

$$\mathcal{P}_t := \frac{SCC_t^r - SCC_t^{T_t^{c,r}}}{SCC_t^{T_t^{c,r}}}. \tag{36}$$

$SCC_t^{T_t^{c,r}}$ is the shadow price of carbon faced by a planner who knows the tipping point location and optimises accordingly, while SCC_t^r is the shadow price faced by an uninformed planner who knows only that the tipping point lies somewhere in $\mathcal{T}$ and uses the robust policy φ^r. $\mathcal{P}_t$ is therefore the percentage increase in the social cost of carbon that the Knightian uncertainty about T^c imposes on the planner, above what they would pay if they faced the worst-case scenario under complete information. Importantly, $\mathcal{P}_t$ is defined ex-ante of the uncertainty about T^c, it is non-negative $\mathcal{P}_t \geq 0$ and $\mathcal{P}_t = 0$ if the uncertainty around the critical threshold T^c is resolved.

Figure 15 shows the simulated uncertainty premium $\mathcal{P}_t$ over time. The uncertainty pre-

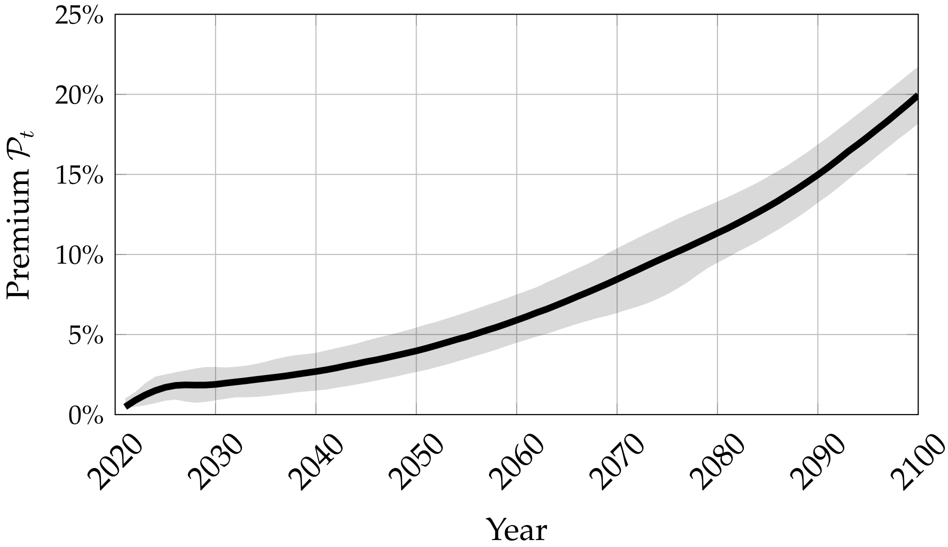

Figure 15: Median and 95% simulation intervals of the uncertainty premium $\mathcal{P}_t$ (36) of 100,000 simulations. The simulation is conditional on temperature T_t not crossing the critical threshold $T_t^{c,r}$.

mium at the beginning of 2020 is between 1% and 2.2%, indicating a modest short-run wedge between the robust and complete information social cost of carbon. Over time, the robust policy becomes increasingly suboptimal, as the implied paths of carbon concentration and temperature diverge from the complete information optimum, and the uncertainty premium $\mathcal{P}_t$ rises to approximately 20% by 2100. This result suggests that the Knightian uncertainty about the tipping point does not justify, in the first couple of decades, a strong premium on the optimal carbon tax placed by a fully informed planner. Yet, as time progresses the welfare cost of this information gap rises rapidly and the marginally suboptimal choices of a robust policy φ^r compound. This suggests that optimal carbon prices should include a rising uncertainty premium to account for the legacy of past robust abatement choices made under unresolved tipping-point uncertainty.

As mentioned in Section 1, the closest approach previously taken in the literature to the estimation of an uncertainty premium on the social cost of carbon is that of Lemoine and Traeger (2016*a*). The authors employ a probabilistic approach to address the Knightian uncertainty about climate tipping points. To arrive at an uncertainty premium Lemoine and Traeger (2016*a*) assume a uniform prior on the critical threshold, thereby inducing a hazard rate on the arrival of a tipping event. In the present paper, I complement this approach by defining a non-probabilistic uncertainty premium which does not require a prior over the critical threshold. In exchange, the estimate found in this paper is possibly over conservative. Despite these differences, and the more recent calibration period of my model, estimates of the uncertainty premium appear consistent and therefore robust to methodological choices.

5 Discussion

This paper studies the impact of Knightian uncertainty about climate tipping points on optimal climate policy. It does so by constructing a non-probabilistic uncertainty premium on the social cost of carbon based on min-max regret policies. The uncertainty premium is between 1% and 2.2% in 2020, but it rises to around 20% by 2100 as the effect of uncertainty compounds. Hence, the ambiguity about tipping thresholds is not a first-order driver of optimal carbon pricing in the short run, yet it becomes increasingly relevant over policy-relevant horizons. Furthermore, I show that the robust policies balance the regret arising from the opportunity costs of abatement, whenever the critical threshold is remote and non-binding, with the regret from the risk of triggering a tipping point when the critical threshold is imminent. It hence acts as an insurance against model misspecification.

The results are consistent with previous estimates of uncertainty premia, however, they do not hinge on assuming the planner is capable of attaching prior probabilities to climate tipping points. This makes the uncertainty premium computed here able to inform policy robustly in the context of climate tipping points, where recent evidence from climate science suggests that both forecasting and learning may be severely constrained.

Two limitations concern both the policy comparison and the climate mechanism. The uncertainty premium is defined relative to a robust policy, whereas real-world climate policy is unlikely to be fully robust and may instead reflect political and institutional frictions. Furthermore, the tipping point considered here is stylised. While it captures the economically relevant feature of self-reinforcing warming, it abstracts from other spillovers and damages that tipping elements may generate. Hence, the quantitative estimates should be interpreted as disciplined measures within a reduced-form framework, rather than as point estimates of the full economic costs of climate tipping points.

Taken together, the evidence in this paper suggests that climate policy ought to account for a rising uncertainty premium in carbon pricing and retain a precautionary orientation (Heal, 2017), even when probabilities are contested.

References

Ackerman, Frank, Elizabeth A. Stanton, and Ramón Bueno. 2013. "Epstein–Zin Utility in DICE: Is Risk Aversion Irrelevant to Climate Policy?" *Environmental and Resource Economics*, 56(1): 73–84. 50

Armstrong McKay, David I., Arie Staal, Jesse F. Abrams, Ricarda Winkelmann, Boris Sakschewski, Sina Loriani, Ingo Fetzer, Sarah E. Cornell, Johan Rockström, and Timothy M. Lenton. 2022. "Exceeding 1.5°C global warming could trigger multiple climate tipping points." *Science*, 377(6611): eabn7950. 3, 5, 10, 49

Athanassoglou, Stergios, and Anastasios Xepapadeas. 2012. "Pollution control with uncertain stock dynamics: When, and how, to be precautious." *Journal of Environmental Economics and Management*, 63(3): 304–320. 6

Baker, Erin, Leon Clarke, and Ekundayo Shittu. 2008. "Technical change and the marginal cost of abatement." *Energy Economics*, 30(6): 2799–2816. 17

Barrage, Lint, and William Nordhaus. 2024. "Policies, projections, and the social cost of carbon: Results from the DICE-2023 model." *Proceedings of the National Academy of Sciences*, 121(13): e2312030121. 52, 53

Barro, Robert J. 2009. "Rare Disasters, Asset Prices, and Welfare Costs." *American Economic Review*, 99(1): 243–264. 4

Ben-Yami, Maya, Andreas Morr, Sebastian Bathiany, and Niklas Boers. 2024. "Uncertainties too large to predict tipping times of major Earth system components from historical data." *Science Advances*, 10(31): eadl4841. 3, 5, 10, 25

Bierkens, Joris, Paul Fearnhead, and Gareth Roberts. 2019. "The Zig-Zag Process and Super-Efficient Sampling for Bayesian Analysis of Big Data." *The Annals of Statistics*, 47(3). arXiv: 1607.03188 [math, stat]. 6, 20, 54, 61

Bondarev, Anton A., and Alfred Greiner. 2018. "Global Warming and Technical Change: Multiple Steady-States and Policy Options." Number: 3112116 Place: Rochester, NY Type: SSRN Scholarly Paper tex.archiveprefix: Social Science Research Network tex.eprint: 3112116. 49

Boulton, Chris A., Lesley C. Allison, and Timothy M. Lenton. 2014. "Early warning signals of Atlantic Meridional Overturning Circulation collapse in a fully coupled climate model." *Nature Communications*, 5(1): 5752–5752. tex.pmcid: 4268699. 10

Burke, Marshall, Solomon Hsiang, and Edward Miguel. 2015. "Global non-linear effect of temperature on economic production." *Nature*, 527(7577): 235–239. 15, 52

Byers, Edward, Volker Krey, Elmar Kriegler, Keywan Riahi, Roberto Schaeffer, Jarmo Kikstra, Robin Lamboll, Zebedee Nicholls, Marit Sandstad, Chris Smith, Kaj van der Wijst, Alaa Al Khourdajie, Franck Lecocq, Joana Portugal-Pereira, Yamina Saheb, Anders Stromman, Harald Winkler, Cornelia Auer, Elina Brutschin, Matthew Gidden, Philip Hackstock, Mathijs Harmsen, Daniel Huppmann, Peter Kolp, Claire Lepault, Jared Lewis, Giacomo Marangoni, Eduardo Müller-Casseres, Ragnhild Skeie, Michaela Werning, Katherine Calvin, Piers Forster, Celine Guivarch, Tomoko Hasegawa, Malte Meinshausen, Glen Peters, Joeri Rogelj, Bjorn Samset, Julia Steinberger, Massimo Tavoni, and Detlef van Vuuren. 2022. "AR6 Scenarios Database." tex.copyright: Restricted Access. 50

Cai, Yongyang, and Thomas S. Lontzek. 2019. "The Social Cost of Carbon with Economic and Climate Risks." *Journal of Political Economy*, 127(6): 2684–2734. 5, 20, 60

Cai, Yongyang, Timothy M. Lenton, and Thomas S. Lontzek. 2016. "Risk of multiple interacting tipping points should encourage rapid CO2 emission reduction." *Nature Climate Change*, 6(5): 520–525. 5

Clarke, Harry R., and William J. Reed. 1994. "Consumption/pollution tradeoffs in an environment vulnerable to pollution-related catastrophic collapse." *Journal of Economic Dynamics and Control*, 18(5): 991–1010. 24

Crost, Benjamin, and Christian P. Traeger. 2013. "Optimal climate policy: Uncertainty versus Monte Carlo." *Economics Letters*, 120(3): 552–558. 50

DeCanio, Stephen J., Charles F. Manski, and Alan H. Sanstad. 2022. "Minimax-regret climate policy with deep uncertainty in climate modeling and intergenerational discounting." *Ecological Economics*, 201: 107552. 26

Dell, Melissa, Benjamin F Jones, and Benjamin A Olken. 2009. "Temperature and Income: Reconciling New Cross-Sectional and Panel Estimates." *American Economic Review*, 99(2): 198–204. 15

Dell, Melissa, Benjamin F Jones, and Benjamin A Olken. 2012. "Temperature Shocks and Economic Growth: Evidence from the Last Half Century." *American Economic Journal: Macroeconomics*, 4(3): 66–95. 15, 16, 52

Deutloff, Jakob, Hermann Held, and Timothy M. Lenton. 2025. "High probability of triggering climate tipping points under current policies modestly amplified by Amazon dieback and permafrost thaw." *Earth System Dynamics*, 16(2): 565–583. 3, 9, 10, 47, 49

DeVries, Tim, Corinne Le Quéré, Oliver Andrews, Sarah Berthet, Judith Hauck, Tatiana Ilyina, Peter Landschützer, Andrew Lenton, Ivan D. Lima, Michael Nowicki, Jörg Schwinger, and Roland Séférian. 2019. "Decadal trends in the ocean carbon sink." *Proceedings of the National Academy of Sciences*, 116(24): 11646–11651. 7

Dietz, Simon, and Bruno Lanz. 2019. "Growth and adaptation to climate change in the long run." IRENE Institute of Economic Research IRENE Working Papers 19-09. 15

Dietz, Simon, and Frank Venmans. 2019. "Cumulative carbon emissions and economic policy: In search of general principles." *Journal of Environmental Economics and Management*, 96: 108–129. 17

Dietz, Simon, Frederick van der Ploeg, Armon Rezai, and Frank Venmans. 2021*a*. "Are Economists Getting Climate Dynamics Right and Does It Matter?" *Journal of the Association of Environmental and Resource Economists*. tex.copyright: © 2021 The Association of Environmental and Resource Economists. All rights reserved. 4, 9, 47, 49

Dietz, Simon, James Rising, Thomas Stoerk, and Gernot Wagner. 2021*b*. "Economic impacts of tipping points in the climate system." *Proceedings of the National Academy of Sciences*, 118(34): e2103081118. 6, 9

Ditlevsen, Peter, and Susanne Ditlevsen. 2023. "Warning of a forthcoming collapse of the Atlantic meridional overturning circulation." *Nature Communications*, 14(1): 4254. 3, 5, 10, 25

Ditlevsen, Peter D., and Sigfus J. Johnsen. 2010. "Tipping points: Early warning and wishful thinking." *Geophysical Research Letters*, 37(19): 2010GL044486. 3, 10

Duffie, Darrell, and Larry G. Epstein. 1992. "Asset Pricing with Stochastic Differential Utility." *Review of Financial Studies*, 5(3): 411–436. 18

Emissions Gap Report 2023: Broken Record – Temperatures hit new highs, yet world fails to cut emissions (again). 2023. *Emissions Gap Report 2023: Broken Record – Temperatures hit new highs, yet world fails to cut emissions (again).* United Nations Environment Programme. 20

Ghil, Michael, and Stephen Childress. 2012. *Topics in geophysical fluid dynamics: Atmospheric dynamics, dynamo theory, and climate dynamics.* Vol. 60, Springer Science & Business Media. 9

Gidden, Matthew J., Keywan Riahi, Steven J. Smith, Shinichiro Fujimori, Gunnar Luderer, Elmar Kriegler, Detlef P. Van Vuuren, Maarten Van Den Berg, Leyang Feng, David Klein, Katherine Calvin, Jonathan C. Doelman, Stefan Frank, Oliver Fricko, Mathijs Harmsen, Tomoko Hasegawa, Petr Havlik, Jérôme Hilaire, Rachel Hoesly, Jill Horing, Alexander Popp, Elke Stehfest, and Kiyoshi Takahashi. 2019. "Global emissions pathways under different socioeconomic scenarios for use in CMIP6: a dataset of harmonized emissions trajectories through the end of the century." *Geoscientific Model Development*, 12(4): 1443–1475. 7, 8, 44, 45

Gilboa, Itzhak. 2009. *Theory of Decision under Uncertainty.* Cambridge:Cambridge University Press. 5

Greiner, Alfred, and Willi Semmler. 2005. "Economic growth and global warming: A model of multiple equilibria and thresholds." *Journal of Economic Behavior & Organization*, 57(4): 430–447. 5

Guivarch, Céline, Thomas Le Gallic, Nico Bauer, Panagiotis Fragkos, Daniel Huppmann, Marc Jaxa-Rozen, Ilkka Keppo, Elmar Kriegler, Tamás Krisztin, Giacomo Marangoni, Steve Pye, Keywan Riahi, Roberto Schaeffer, Massimo Tavoni, Evelina Trutnevyte, Detlef van Vuuren, and Fabian Wagner. 2022. "Using large ensembles of climate change mitigation scenarios for robust insights." *Nature Climate Change*, 12(5): 428–435. tex.copyright: 2022 Springer Nature Limited. 47

Hambel, Christoph, Holger Kraft, and Eduardo Schwartz. 2021. "Optimal carbon abatement in a stochastic equilibrium model with climate change." *European Economic Review*, 132: 103642. 4, 7, 9, 14, 15, 17, 19, 46, 50, 54

Heal, Geoffrey. 2017. "The Economics of the Climate." *Journal of Economic Literature*, 55(3): 1046–1063. 33

Heal, Geoffrey, and Antony Millner. 2014. "Reflections: Uncertainty and Decision Making in Climate Change Economics." *Review of Environmental Economics and Policy*, 8(1): 120–137. 6, 26

Howard, Peter H., and Thomas Sterner. 2017. "Few and not so far between: a meta-analysis of climate damage estimates." *Environmental and Resource Economics*, 68(1): 197–225. 52

IPCC. 2023*a*. *Climate Change 2021 – The Physical Science Basis: Working Group I Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change.* . 1 ed., Cambridge University Press. 44

IPCC, ed. 2023*b*. *Climate Change 2022 - Mitigation of Climate Change: Working Group III Contribution to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change.* . 1 ed., Cambridge University Press. 17, 50

Johnson, Steven G. 2007. "The NLopt nonlinear-optimization package." 62

Kalkuhl, Matthias, and Leonie Wenz. 2020. "The impact of climate conditions on economic production. Evidence from a global panel of regions." *Journal of Environmental Economics and Management*, 103: 102360. 15, 16, 52

Kamien, Morton I., and Nancy L. Schwartz. 1971. "Sufficient conditions in optimal control theory." *Journal of Economic Theory*, 3(2): 207–214. 4

Kraft, Holger, and Frank Thomas Seifried. 2014. "Stochastic differential utility as the continuous-time limit of recursive utility." *Journal of Economic Theory*, 151: 528–550. 59

Kushner, Harold J. 2007. "Numerical Approximations for Nonzero-Sum Stochastic Differential Games." *SIAM Journal on Control and Optimization*, 46(6): 1942–1971. 54, 58

Kushner, Harold J., and Paul Dupuis. 2001. *Numerical Methods for Stochastic Control Problems in Continuous Time.* Vol. 24 of *Stochastic Modelling and Applied Probability*, New York, NY:Springer New York. 6, 20, 54, 58, 60

Leach, Nicholas J., Stuart Jenkins, Zebedee Nicholls, Christopher J. Smith, John Lynch, Michelle Cain, Tristram Walsh, Bill Wu, Junichi Tsutsui, and Myles R. Allen. 2021. "FaIRv2.0.0: a generalized impulse response model for climate uncertainty and future scenario exploration." *Geoscientific Model Development*, 14(5): 3007–3036. 9, 47

Le Kama, Alain Ayong, and Katheline Schubert. 2004. "Growth, Environment and Uncertain Future Preferences." *Environmental and Resource Economics*, 28(1): 31–53. 19

Lemoine, Derek, and Christian P. Traeger. 2016*a*. "Ambiguous tipping points." *Journal of Economic Behavior & Organization*, 132: 5–18. 5, 6, 32

Lemoine, Derek, and Christian P. Traeger. 2016*b*. "Economics of tipping the climate dominoes." *Nature Climate Change*, 6(5): 514–519. 5

Lemoine, Derek, and Christian Traeger. 2014. "Watch Your Step: Optimal Policy in a Tipping Climate." *American Economic Journal: Economic Policy*, 6(1): 137–166. 5, 6

Le Quéré, Corinne, Christian Rödenbeck, Erik T. Buitenhuis, Thomas J. Conway, Ray Langenfelds, Antony Gomez, Casper Labuschagne, Michel Ramonet, Takakiyo Nakazawa, Nicolas Metzl, Nathan Gillett, and Martin Heimann. 2007. "Saturation of the Southern Ocean CO2 Sink Due to Recent Climate Change." *Science*, 316(5832): 1735–1738. 7

Li, Chuan-Zhong, Anne-Sophie Crépin, and Therese Lindahl. 2024. "The Economics of Tipping Points: Some Recent Modeling and Experimental Advances." *International Review of Environmental and Resource Economics*, 18(4): 385–442. 4, 5

Lin, Xu, and Sweder Van Wijnbergen. 2023. "The Social Cost of Carbon under Climate Volatility Risk." Tinbergen Institute Tinbergen Institute Discussion Papers 23-032/IV. 4

Lontzek, Thomas S., Yongyang Cai, Kenneth L. Judd, and Timothy M. Lenton. 2015. "Stochastic integrated assessment of climate tipping points indicates the need for strict climate policy." *Nature Climate Change*, 5(5): 441–444. 4, 50

McGuffie, Kendal, and Ann Henderson-Sellers. 2014. *The Climate Modelling Primer.* . 4. ed ed., Chicester:Wiley Blackwell. 5, 9, 10, 11

Millner, Antony, Simon Dietz, and Geoffrey Heal. 2013. "Scientific Ambiguity and Climate Policy." *Environmental and Resource Economics*, 55(1): 21–46. 6

Mogensen, Patrick Kofod, and Asbjørn Nilsen Riseth. 2018. "Optim: A mathematical optimization package for Julia." *Journal of Open Source Software*, 3(24): 615. 61

Mäler, Karl-Göran, Anastasios Xepapadeas, and Aart de Zeeuw. 2003. "The Economics of Shallow Lakes." *Environmental and Resource Economics*, 26(4): 603–624. 5

Nordhaus, William D. 1992. "An optimal transition path for controlling greenhouse gases." *Science*, 258(5086): 1315–1319. 17

Nordhaus, William D. 2014. "Estimates of the Social Cost of Carbon: Concepts and Results from the DICE-2013R Model and Alternative Approaches." *Journal of the Association of Environmental and Resource Economists*, 1(1): 273–312. 50

Nordhaus, William D. 2017. "Revisiting the social cost of carbon." *Proceedings of the National Academy of Sciences of the United States of America*, 114(7): 1518–1523. 17

Nordhaus, William D. 2019. "Economics of the disintegration of the Greenland ice sheet." *Proceedings of the National Academy of Sciences*, 116(25): 12261–12269. 5

Nordhaus, William D., and Andrew Moffat. 2017. "A Survey of Global Impacts of Climate Change: Replication, Survey Methods, and a Statistical Analysis." National Bureau of Economic Research w23646, Cambridge, MA. 52

Nævdal, Eric, and Michael Oppenheimer. 2007. "The economics of the thermohaline circulation—A problem with multiple thresholds of unknown locations." *Resource and Energy Economics*, 29(4): 262–283. 4

Oberkampf, William L., and Christopher J. Roy. 2010. *Verification and Validation in Scientific Computing*. . 1 ed., Cambridge University Press. 60

Olijslagers, Stan, and Sweder Van Wijnbergen. 2019. "Discounting the Future: On Climate Change, Ambiguity Aversion and Epstein-Zin Preferences." *SSRN Electronic Journal*. 4

Parry, Isobel M., Paul D. L. Ritchie, and Peter M. Cox. 2022. "Evidence of localised Amazon rainforest dieback in CMIP6 models." *Earth System Dynamics*, 13(4): 1667–1675. 10

Pindyck, Robert S., and Neng Wang. 2013. "The Economic and Policy Consequences of Catastrophes." *American Economic Journal: Economic Policy*, 5(4): 306–339. 4, 14, 19, 54

Rackauckas, Christopher, and Qing Nie. 2017. "Adaptive methods for stochastic differential equations via natural embeddings and rejection sampling with memory." *Discrete and continuous dynamical systems. Series B*, 22(7): 2731. 61

Rackauckas, Christopher, Yingbo Ma, Julius Martensen, Collin Warner, Kirill Zubov, Rohit Supekar, Dominic Skinner, and Ali Ramadhan. 2020. "Universal differential equations for scientific machine learning." *arXiv preprint arXiv:2001.04385*. arXiv: 2001.04385. 44

Rezai, Armon, and Frederick Van Der Ploeg. 2017. "Climate policies under climate model uncertainty: Max-min and min-max regret." *Energy Economics*, 68: 4–16. 26

Rietkerk, Max, Vanessa Skiba, Els Weinans, Raphaël Hébert, and Thomas Laepple. 2025. "Ambiguity of early warning signals for climate tipping points." *Nature Climate Change*, 15(5): 479–488. 3, 5, 11, 25

Rudik, Ivan. 2020. "Optimal Climate Policy When Damages are Unknown." *American Economic Journal: Economic Policy*, 12(2): 340–373. 6

Savage, L. J. 1951. "The Theory of Statistical Decision." *Journal of the American Statistical Association*, 46(253): 55–67. 26

Seaver Wang, A. Foster, E. A. Lenz, J. Kessler, J. Stroeve, L. Anderson, M. Turetsky, R. Betts, Sijia Zou, W. Liu, W. Boos, and Z. Hausfather. 2023. "Mechanisms and Impacts of Earth System Tipping Elements." *Reviews of Geophysics*. 3, 5, 10

Shi, Hao, Hanqin Tian, Naiqing Pan, Christopher P. O. Reyer, Philippe Ciais, Jinfeng Chang, Matthew Forrest, Katja Frieler, Bojie Fu, Anne Gädeke, Thomas Hickler, Akihiko Ito, Sebastian Ostberg, Shufen Pan, Miodrag Stevanović, and Jia Yang. 2021. "Saturation of Global Terrestrial Carbon Sink Under a High Warming Scenario." *Global Biogeochemical Cycles*, 35(10): e2020GB006800. 7

Skiba, A. K. 1978. "Optimal Growth with a Convex-Concave Production Function." *Econometrica : journal of the Econometric Society*, 46(3): 527. JSTOR: 1914229. 5

Smith, Christopher J., Piers M. Forster, Myles Allen, Nicholas Leach, Richard J. Millar, Giovanni A. Passerello, and Leighton A. Regayre. 2017. "FAIR v1.1: A simple emissions-based impulse response and carbon cycle model." 5

Smolders, Emma J. V., René M. van Westen, and Henk A. Dijkstra. 2024. "Probability Estimates of a 21st Century AMOC Collapse." arXiv: 2406.11738 [physics] Number: arXiv:2406.11738. 3, 25

Svanberg, Krister. 2002. "A class of globally convergent optimization methods based on conservative convex separable approximations." *SIAM Journal on Optimization*, 12: 555–573. 62

Tol, Richard S.J. 2024. "A meta-analysis of the total economic impact of climate change." *Energy Policy*, 185: 113922. 52

Traeger, Christian P. 2014. "Why uncertainty matters: discounting under intertemporal risk aversion and ambiguity." *Economic Theory*, 56(3): 627–664. 6

Tsur, Yacov, and Amos Zemel. 1996. "Accounting for global warming risks: Resource management under event uncertainty." *Journal of Economic Dynamics and Control*, 20(6-7): 1289–1305. 4

Turetsky, Merritt R., Benjamin W. Abbott, Miriam C. Jones, Katey Walter Anthony, David Olefeldt, Edward A. G. Schuur, Charles Koven, A. David McGuire, Guido Grosse, Peter Kuhry, Gustaf Hugelius, David M. Lawrence, Carolyn Gibson, and A. Britta K. Sannel. 2019. "Permafrost collapse is accelerating carbon release." *Nature*, 569(7754): 32–34. 10

Van den Bremer, Ton S., and Frederick Van der Ploeg. 2021. "The Risk-Adjusted Carbon Price." *American Economic Review*, 111(9): 2782–2810. 4

Van der Ploeg, Frederick, and Aart De Zeeuw. 2018. "Climate Tipping and Economic Growth: Precautionary Capital and the Price of Carbon." *Journal of the European Economic Association*, 16(5): 1577–1617. 4

Van Westen, René M., Michael Kliphuis, and Henk A. Dijkstra. 2024. "Physics-based early warning signal shows that AMOC is on tipping course." *Science Advances*, 10(6): eadk1189. 10

Wagener, Florian. 2013. "Regime Shifts: Early Warnings." In *Encyclopedia of Energy, Natural Resource, and Environmental Economics*. 349–359. Elsevier. 5, 11

Wagener, Florian. 2015. "Economics of Environmental Regime Shifts." In *The Oxford Handbook of the Macroeconomics of Global Warming*. , ed. Lucas Bernard and Willi Semmler, 0. Oxford University Press. 5

Weitzman, Martin L. 2012. "GHG Targets as Insurance against Catastrophic Climate Damages." *Journal of Public Economic Theory*, 14(2): 221–244. 53

Weitzman, Martin L. 2014. "Fat Tails and the Social Cost of Carbon." *American Economic Review*, 104(5): 544–546. 16

A Multiplicity of Climate Equilibria

This section derives the analytical conditions under which the climate block admits multiple temperature equilibria for a fixed atmospheric concentration level. The deterministic component of the temperature dynamics (10) is given by

$$\epsilon \frac{\mathrm{d}T_t}{\mathrm{d}t} = r^{T^c}(T_t) + g(M), \tag{37}$$

where, $r^{T^c}(T)$ is given in (8) and $g(M)$ in (9).

For a fixed concentration level M, equilibrium temperatures $\bar{T}$ solve

$$F(\bar{T}, M) = r^{T^c}(\bar{T}) + g(M) = 0. \tag{38}$$

Multiple equilibria occur when (38) admits three roots: two locally stable equilibria separated by one unstable equilibrium. The boundaries of this multiplicity region are fold (saddle-node) points (T_*, M_*), defined by the system

$$F(T_*; M_*) = 0, \tag{39}$$

$$\partial_T F(T_*; M_*) = 0. \tag{40}$$

Using the specification above, (40) is

$$\left(\lambda^{T^c}\right)'(T_*) - 4\eta\sigma T_*^3 = 0, \tag{41}$$

and (39) implies

$$M_* = M^p \exp\left(-\frac{S_0 + \lambda^{T^c}(T_*) - \eta\sigma T_*^4 + G_0}{G_1}\right). \tag{42}$$

Hence, if (41) admits two solutions $T_- < T_+$, the corresponding concentrations M_- and M_+ obtained from (42) delimit the multiplicity interval: for $M \in (M_-, M_+)$ there are three equilibria, while for $M \notin [M_-, M_+]$ the equilibrium is unique.

Local stability is determined by the sign of $\partial_T F$ at each equilibrium T^e: equilibrium is locally stable if

$$\partial_T F(T^e; M) = \left(\lambda^{T^c}\right)'(T^e) - 4\eta\sigma(T^e)^3 < 0, \tag{43}$$

and unstable if the inequality is reversed.

B Calibration and Parameters

B.1 Climate Model

This section discusses the calibration of the climate model. The optimisation of differential equations parameters is done via the `SciMLSensitivity.jl` and `Optim` packages (Rackauckas et al., 2020). The baseline year $t = 0$ of the calibration is chosen to be 2020. The calibration horizon is of $\tau = 80$ years, such that the calibration matches variables through 2100. The remaining period between 2100 and 2150 is used as an out-of-sample validation for the calibration.

B.1.1 Greenhouse Gasses Emissions under No-Policy

I assume the emission of greenhouse gasses (GHGs) in the no-policy (np) scenario follows the median path in the SSP5-8.5 scenario, sourced from the CMIP6 GHG and aerosol emission project (Gidden et al., 2019). First, I extract median paths $\hat{M}^{np}_{c,t}$ in p.p.m. of five major GHGs

$$c \in \{\, \mathrm{CO_2},\ \mathrm{CH_4},\ \mathrm{N_2O},\ \mathrm{NF_3},\ \mathrm{SF_6}\}.$$

Finally, I construct the no-policy path of the $\mathrm{CO_2e}$ atmospheric concentration as

$$\hat{M}^{np}_t = \hat{M}^{np}_{\mathrm{CO_2},t} + \sum_c \mathrm{GWP}_c \times \hat{M}^{np}_{c,t}, \tag{44}$$

where GWP_c is the Global Warming Potential of the GHG c, as provided in the AR6, Working Group I from the IPCC (2023*a*). These are summarised in table 2. Figure 16 illustrates the

GHG c	GWP
$\mathrm{CH_4}$	29.8
$\mathrm{N_2O}$	273
$\mathrm{NF_3}$	17400
$\mathrm{SF_6}$	24300

Table 2: Global Warming Potential used in the calibration.

fraction of atmospheric $\mathrm{CO_2}$ concentration $\hat{M}^{np}_{t,\mathrm{CO_2}}$ to the fraction of the $\mathrm{CO_2e}$ atmospheric concentration $\hat{M}^{np}_t$ (44). Given this calibration, around 18% of radiative forcing in 2020 is due to non- $\mathrm{CO_2}$ GHGs. Later in the century, $\mathrm{CO_2}$ becomes a larger contributor of radiative forcing, as the emissions of all GHGs tend to fall and $\mathrm{CO_2}$ has a longer half-life of other GHGs.

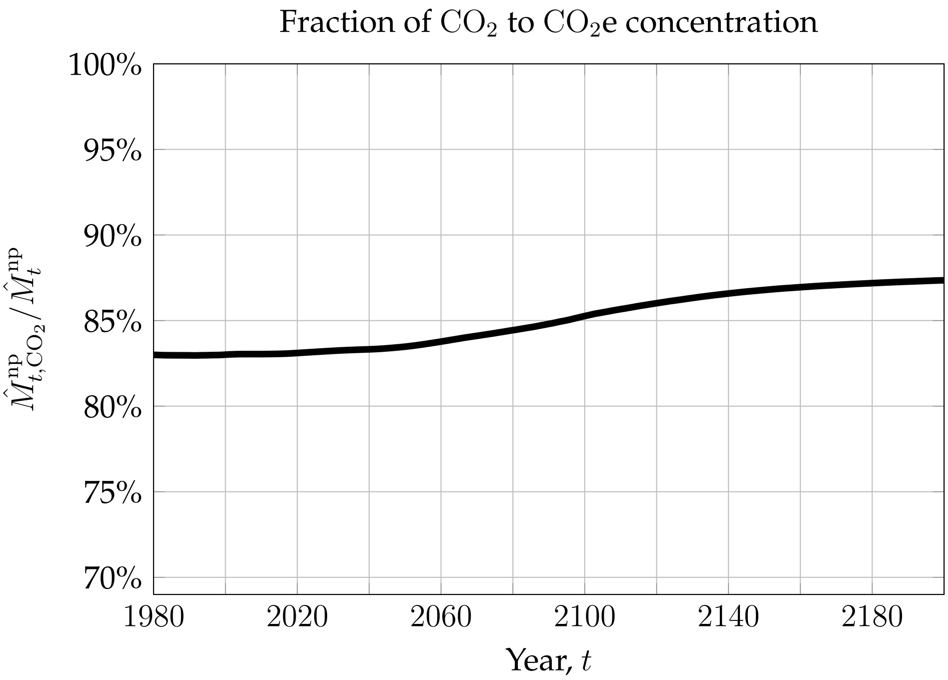

Figure 16: Fraction of CO_2 to CO_2e concentration, $\hat{M}^{np}_{t,\,\mathrm{CO}_2}/\hat{M}^{np}_t$

For tractability, I further estimate the growth rate γ^{np}_t (2) of CO_2e concentration M^{np}_t by assuming it to be a polynomial function of time as

$$\gamma^{np}_t = \sum_{n=0}^{k} c_n t^n \tag{45}$$

and estimate the parameters $\{c_n\}_n$ and the order k to minimise the error between the calibrated CO_2e concentration $M^{np}_t = M_0 e^{\int_0^\tau \gamma^{np}_t \mathrm{d}t}$ and the path of CO_2e concentration $\hat{M}^{np}_t$ implied in Gidden et al. (2019), over the horizon τ, and a penalty term for the order k, that is,

$$(k, \{c_n\}_n) = \arg\min_{k,\{c_n\}_n} \int_0^\tau |M^{np}_t - \hat{M}^{np}_t|^2 \mathrm{d}t + \lambda k^2. \tag{46}$$

The resulting order k and coefficients $\{c_n\}_n$ are displayed in Table 3 and the calibrated γ^{np}_t and M^{np}_t are illustrated in Figure 1. The calibration's maximal absolute error $\max_t |M^{np}_t - \hat{M}^{np}_t|$ is

Parameter	Calibrated Value
k	4
c_0	-0.0001730
c_1	-0.09061
c_2	0.0001288
c_3	-6.10×10^{-8}
c_4	9.61×10^{-12}

Table 3: Calibrated parameters k and $\{c_n\}_n$ for the growth rate γ^{np}_t (45) of M^{np}_t.

0.009785412 p.p.m..

B.1.2 Decay Rate of Carbon

Following Hambel, Kraft and Schwartz (2021), I make an assumption on the decay rate of carbon. The carbon decay δ_m, as a function of the stored carbon M_t is assumed to take the functional form

$$\delta_m(M_t) = a_\delta e^{-\left(\frac{d_\delta M_t - b_\delta}{c_\delta}\right)^2}, \tag{47}$$

for parameters $a_\delta, b_\delta, c_\delta$ and d_δ.

I calibrate the concentration-dependent decay term using the gap between the increase in concentration under the no policy, $M_t^{np} - M_0^{np}$ and the cumulative emissions $\int_0^t E_s^{np} \mathrm{d}s$, as illustrated in Figure 17. The implied decay is then

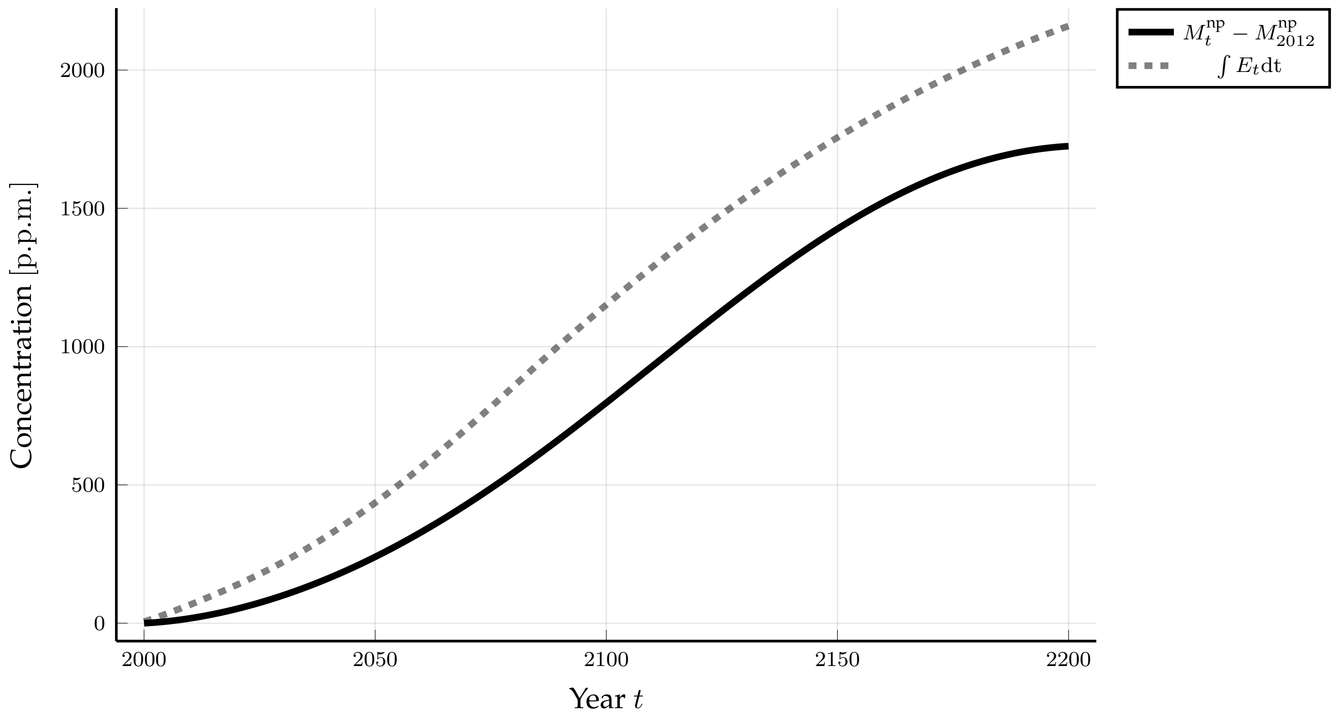

Figure 17

$$\hat{\delta}_t = \frac{E_t^{np}}{M_t^{np}} - \gamma_t. \tag{48}$$

Parameters are chosen to match the implied decay profile by minimizing the absolute discrepancy across the calibration sample

$$\min_{a_\delta, b_\delta, c_\delta, d_\delta} \int_\tau^0 \left| \hat{\delta}_t - \delta_m(M_t) \right| \mathrm{d}t, \tag{49}$$

Using this setup, under a no-policy scenario, the calibrated parameters are given in Table 4.

The in-sample error is of 7.86×10^{-5} and the out-of-sample error is of 8.87×10^{-3}.

Parameter	Calibrated value
a_δ	0.013
b_δ	0
c_δ	296.25
d_δ	0.398

Table 4: Calibrated parameters of the carbon decay function $\delta_m(M_t)$, rounded to three significant digits.

B.1.3 Temperature in Absence of Tipping Elements

I calibrate GHG forcing g (9) to match the mapping between GHG and temperature in the FaIRv2 model developed by Leach et al. (2021). I choose this model to be consistent with the dynamics in Deutloff, Held and Lenton (2025), which is then used to calibrate the temperature impact of tipping elements. I simulate the path of $\mathrm{CO_2e}$ carbon concentration M_t^{np} under the no-policy scenario np using the FaIRv2 model. Third, I generate a probabilistic projection of temperature deviations from pre-industrial levels under M_t^{np} and extract the median temperature path $\hat{T}_t^{np}$. Finally, assuming no positive-feedback in temperature (8), that is, $\lambda(T_t) \equiv \lambda_1$, I calibrate the GHGs forcing parameters G_0, G_1, in g (9) to minimise the resulting median temperature path simulated under the temperature dynamics (10), that is,

$$(G_0, G_1) = \arg \min_{G_0, G_1} \int_0^\tau |T_t^{np} - \hat{T}_t^{np}|^2 \mathrm{d}t. \tag{50}$$

The calibration targets the median path of temperature from FaIRv2 as the uncertainty in the FaIRv2 projections of temperature is mostly due to uncertainty in models and in emissions, rather than to yearly temperature variability. See Guivarch et al. (2022) for a detailed discussion. The calibrated parameters are reported in Table 5 and the resulting median temperature dynamics T_t^{np} are shown in Figure 18. The calibration maximal absolute error $\max_t |T_t^{np} - \hat{T}_t^{np}|$

	Calibrated Value	Description
G_0	150.424	Intercept
G_1	22.687	Slope

Table 5: Calibrated parameters G_0 and G_1 for the $\mathrm{CO_2e}$ forcing (9).

is 0.05705 °C in the estimation period and 0.23766 °C out of sample.

I further validate the calibration by simulating the impulse response of temperature to a one-off increase of $\mathrm{CO_2e}$ concentration of 47 p.p.m. and compute the time it takes median temperature to reach half way to its equilibrium level, defined as "half-life" in Dietz et al. (2021*a*). The median half-life in the calibrated model is approximately 6 months,

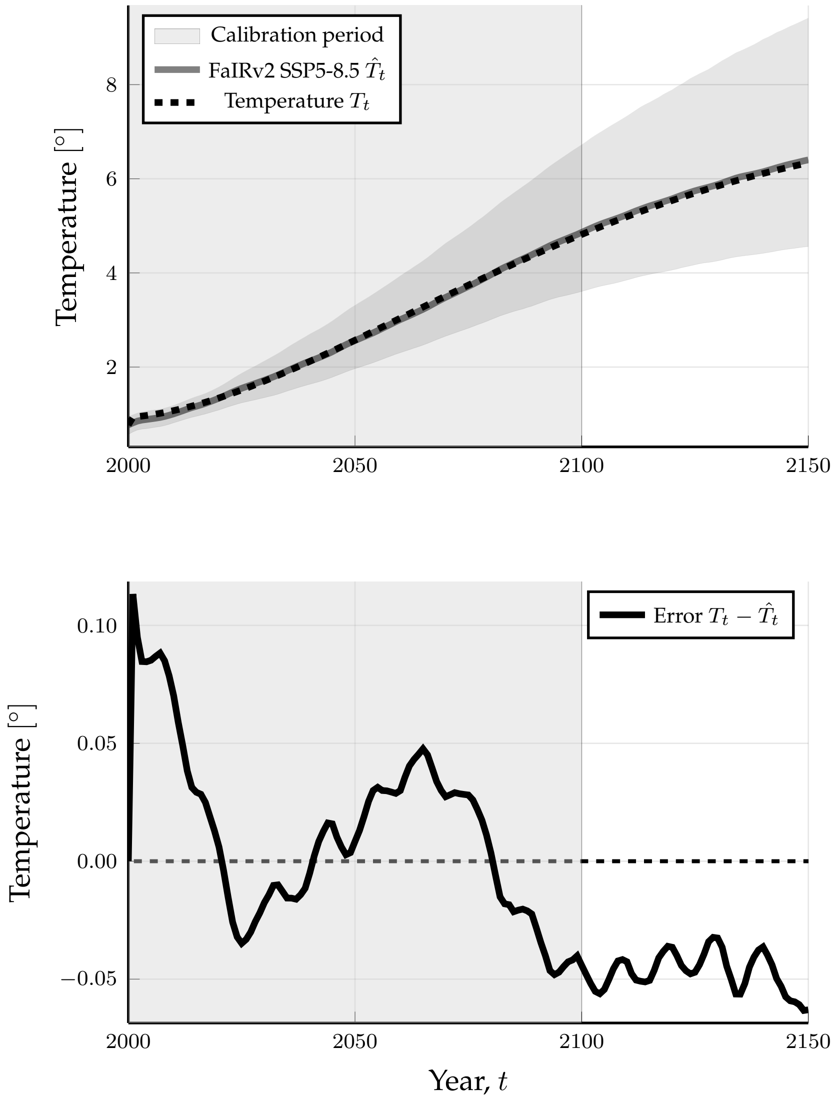

Figure 18: Median calibrated temperature T_t^{np} path and the target FaIRv2 temperature projections $\hat{T}_t^{np}$ with 95% confidence intervals.

which is faster than the approximately 3 years FaIRv2 model. Despite this large error, the model still performs better than many models commonly employed in the IAM literature (e.g. DICE2016), which display a very sluggish response of temperature to $\mathrm{CO_2e}$ (Dietz et al., 2021*a*).

B.1.4 Temperature Impact of Tipping Elements

The impact of temperature feedbacks on temperature is calibrated from Deutloff, Held and Lenton (2025), hereafter DHL. The critical temperatures of the feedback mechanisms, T^c in the context of my model, are chosen by DHL to be the central estimates of Armstrong McKay et al. (2022). Given the wide uncertainty in these estimates, here I consider a more robust, albeit prudent approach, by not calibrating T^c and computing the regret for various values of the critical temperature. Hence, I rely on DHL to match only the impact on temperature, and not the timing, of the feedback mechanisms. The transition function $L(T_t - T^c)$ of the feedback mechanism (7) takes the form[7]

$$L(T_t - T^c) = \frac{1}{1 + \exp\left(L_1\left(T_t - T^c\right) + L_0\right)}. \tag{51}$$

I calibrate the parameters $\Delta\lambda$, L_0, and L_1 by matching the additional warming at the end of the century between the DHL model with temperature feedbacks ("coupled") and without temperature feedbacks ("uncoupled"). More formally, introduce the radiative forcing

$$r^l(T_t) \coloneqq S_0\left(1 - \lambda_1\right) - \eta\sigma T_t^4 \tag{52}$$

in absence of positive temperature feedbacks and $T_t^{l,np}$ the temperature path satisfying (10) with radiative forcing (52) under the np scenario. Then, the impact of temperature feedbacks on the end-of-century is defined as

$$\Delta T^{np} = T_{2100}^{np} - T_{2100}^{l,np}. \tag{53}$$

Then the parameters $\Delta\lambda$, L_0, and L_1 are chosen to minimise the square distance of the median, 5% and 95% quantiles between ΔT^{np} and the DHL additional warming $\hat{\Delta T}^{np}$. The results of the calibration are provided in Table 6.

[7] See Bondarev and Greiner (2018) for a discussion on the effect of an alternative specification on the results of the optimisation problem.

	Calibrated Value	Description
L_0	3.15	Location Parameter
L_1	3.5 [°C^{-1}]	Speed Parameter
λ_1	0.31	Initial level of feedback
ΔS	0.01545 [Wm^{-2}]	Magnitude of feedback

Table 6: Calibration of feedback mechanism (51).

B.1.5 Other Climate Parameters

The remaining parameters of the climate model are reported in Table 7.

Climate		
T_0	1.34 [°C]	Initial temperature
M_0	558.75 [p.p.m.]	Initial CO_2e concentration
M^{p}	383.15 [p.p.m.]	Pre-industrial CO_2e concentration
σ_T	1.5844 [y$^{-1/2}$]	Volatility of temperature
S_0	340.5 [W m^{-2}]	Mean solar radiation
ϵ	15.844 [J m^{-2} K y]	Heat capacity of the ocean
η	5.67×10^{-8}	Stefan-Boltzmann constant

Table 7: Remaining parameters of the climate model.

B.2 Economy

B.2.1 Preferences

The following Table 8 illustrates the preferences parameters used throughout the paper. There is no consensus in the literature on preference parameters. In line with previous literature focusing on recursive preferences, I set relative risk aversion $\theta = 10$ (Ackerman, Stanton and Bueno, 2013; Crost and Traeger, 2013; Lontzek et al., 2015) and the time preference parameter $\rho = 1.5\%$ (Nordhaus, 2014). Following the discussion in Hambel, Kraft and Schwartz (2021), I choose $\psi = 1$.

Preferences		
ρ	1.5%	Time preference
θ	10	Relative risk aversion
ψ	1	Elasticity of intertemporal substitution

Table 8: Parameters of preferences.

B.2.2 Marginal Abatement Curves

The function β_t' (19) is calibrated to match the implied marginal abatement curves in the Sixth Assessment Report of the IPCC (2023*b*, Figure 3.33). The data is sourced from Byers et al. (2022). In the model, the marginal abatement cost $\mathrm{MAC}_t := -\frac{\partial B_t}{\partial E_t}$ can be linked to β' via a straightforward calculation, that is,

$$\beta'_t = \mathrm{MAC}_t \frac{E^{\mathrm{np}}_t}{Y_t}. \tag{54}$$

For each SSP scenario and each IAM m in the dataset, I extract carbon prices, which are assumed to be equal to marginal abatement costs $\mathrm{MAC}_{m,t}$, emission $E_{m,t}$, baseline emissions $E^{\mathrm{np}}_{m,t}$, and output $Y_{m,t}$. Then, I compute the implied

$$\beta'_{m,t} = \mathrm{MAC}_{m,t} \frac{E^{\mathrm{np}}_{m,t}}{Y_{m,t}} \tag{55}$$

and

$$\varepsilon_{m,t} = 1 - \frac{E_{m,t}}{E^{\mathrm{np}}_{m,t}}. \tag{56}$$

Then, I compute for $t = 0$ the power b that minimises the mean-squared error $\beta'_{m,0} = \log(b\omega_0) + (b-1)\log \varepsilon_{m,0}$. Finally, I estimate via linear regression, for each model m, the coefficients $\alpha_{m,0}$ and $\alpha_{m,1}$ of

$$\log \beta'_{m,t} - \log b - (b-1)\log \varepsilon_{m,t} = \alpha_{m,0} + \alpha_{m,1} t, \tag{57}$$

noticing that these can be related, via the abatement technology parameter ω_t (20), to the parameters of interest

$$\omega_0 = \log(\alpha_{m,0}), \tag{58}$$

$$\rho_\omega = -\alpha_{m,1}. \tag{59}$$

This procedure yields the estimates in Table 9. In the paper, the estimates used are those of

Model m	ω_0		ρ_ω	
IMAGE	0.0532	(0.003298)	0.0045	(0.000023)
WITCH-GLOBIOM	0.0633	(0.002079)	-0.0215	(0.000014)
AIM/CGE	0.0560	(0.002265)	0.0071	(0.000017)

Table 9: Estimated β' parameters for each model m. Standard errors in parentheses.

the IMAGE model, that is,

$$\omega_0 = 10.64\%, \rho_\omega = 0.45\% \text{ and } b = 2.8. \tag{60}$$

B.2.3 Damage Function

Most integrated assessment models, including the one presented in this paper, summarise the effect of an increase in global average temperatures as a simple damage function d. Despite its convenience, there is no broad consensus on whether damages ought to be a function of temperature T_t or temperature changes $\mathrm{d}T_t$, the form this function should take, and whether the damages should affect output Y_t or output growth $\mathrm{d}Y_t$. For an overview on this debate, I refer the reader to the surveys by Howard and Sterner (2017), Nordhaus and Moffat (2017), and Tol (2024).

In this paper, I follow recent evidence on the effect of temperature T_t on productivity growth $\mathrm{d}A_t$ (Burke, Hsiang and Miguel, 2015; Dell, Jones and Olken, 2012; Kalkuhl and Wenz, 2020), and assume that climate damages d_t at time t lower the growth rate of productivity

$$\frac{\mathrm{d}A_t}{\mathrm{d}t} = \varrho - d_t. \tag{61}$$

I employ a quadratic damage function (14) and the calibration by Kalkuhl and Wenz (2020, Table 4, Section 6.2). The authors estimate the quadratic cumulative damages of warming since pre-industrial levels on cumulative loss in GDP growth, assuming quadratic damages (14), equivalent to the formulation used in Burke, Hsiang and Miguel (2015). The estimated parameter is $\xi = 0.0709\%$.

A broad set of integrated assessment models, most notably the DICE model, with its latest iteration in Barrage and Nordhaus (2024), assume temperature T_t to decrease output levels rather than growth. To compare the two approaches, let $\tilde{Y}_t$ denote the counterfactual output path under the same non-climate growth components, but in absence of climate damages, and define level damages by $Y_t = (1 - D_t)\tilde{Y}_t$. Under the growth specification in this paper, productivity satisfies

$$A_t = A_0 \exp\left(\int_0^t \left(\varrho - d(T_s)\right) \mathrm{d}s\right), \qquad A_t^0 = A_0 e^{\varrho t}. \tag{62}$$

Hence, holding fixed the remaining components of output growth, the ratio of damaged to undamaged output is given by

$$\frac{Y_t}{\tilde{Y}_t} = \frac{A_t}{A_t^0} = \exp\left(-\int_0^t d(T_s)\mathrm{d}s\right), \tag{63}$$

so that the growth-damage specification is equivalent to the level wedge

$$D_t = 1 - \exp\left(-\int_0^t d(T_s)\mathrm{d}s\right). \tag{64}$$

Therefore, the level damage implied by a growth specification generally depends on the full history of temperature, not only on current temperature T_t. For comparison purposes, I compute implied level damages D_t from growth damages under the following assumptions. First, the growth rate of atmospheric $\mathrm{CO_2e}$ concentration is constant at its initial level, $\bar{\gamma} \equiv \gamma_0$, such that $M_t = M^p e^{\bar{\gamma}t}$. Second, the climate is deterministic and there is no tipping point. Third, temperature is in equilibrium, such that

$$T_t = \left(\frac{S_0\left(1-\lambda_1\right) + G_1\bar{\gamma}t}{\eta\sigma}\right)^{\frac{1}{4}}. \tag{65}$$

Then, I numerically compute (64) and report the pair (D_t, T_t). Figure 19 compares implied level damages for two concentration-growth paths, $\bar{\gamma} = \gamma_0$ and $\bar{\gamma} = 2\gamma_0$. As growth damages

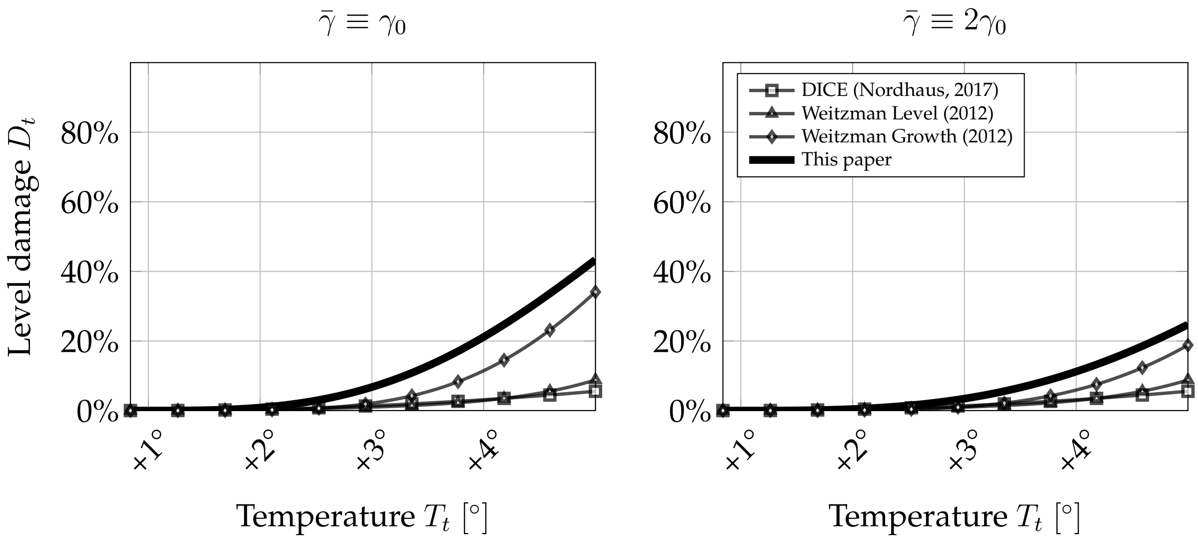

Figure 19: Implied level damages from the growth-damage specification in this paper, compared with the level and growth damage functions in Weitzman (2012) and the DICE-2023 level damages (Barrage and Nordhaus, 2024). Both panels assume equilibrium temperature and deterministic climate dynamics, but differ in the growth rate of atmospheric $\mathrm{CO_2e}$ concentration. The contrast illustrates that there is no unique joint decomposition of level damages into a contemporaneous temperature effect and an intertemporal accumulation effect.

cumulate over time, their implied level damages can be much larger than directly specified level-damage functions. However, Figure 19 shows that this comparison is not unique: changing $\bar{\gamma}$ changes how quickly the economy moves along the temperature path and therefore how long growth damages accumulate at each temperature level. Consequently, the same observed level-damage profile can be rationalised by different combinations of instantaneous temperature effects and accumulation-over-time effects. Particularly, observe that increasing the growth rate $\bar{\gamma}$ paradoxically reduces the implied level damages as given temperature levels are reached earlier. Because of this, it is hard to construct a principled comparison between growth and level damage functions.

B.2.4 Macroeconomy

For the remaining calibration of the economy (Table 10), I follow the calibration suggested by Hambel, Kraft and Schwartz (2021, Section 3). I first calibrate the expected growth rate $\varrho + \phi(\chi_t)$ of output (22) in absence of abatement measures to match the DICE model through 2200. As in Hambel, Kraft and Schwartz (2021), as DICE is deterministic, I match the average simulation of my model to the DICE output. Following Pindyck and Wang (2013), the initial level of productivity is set at $A(0) = 0.113$.

Economy		
ϱ	0.09% $[\mathrm{y}^{-1}]$	Growth of TFP
κ	6.32% $[\mathrm{y}^{-1}]$	Adjustment costs of abatement technology
δ_k	0.0116 $[\mathrm{y}^{-1}]$	Initial depreciation rate of capital
A_0	0.113	Initial TFP
Y_0	75.8 $[\mathrm{trUS\$/y}]$	Initial GDP
σ_k	0.0162 $[\mathrm{y}^{-1/2}]$	Variance of GDP
τ	500 $[\mathrm{y}]$	Steady state horizon

Table 10: Parameters of the economic model.

C Numerical Solution of Climate-Economy Model

This appendix deals with the solution of the maximisation problem (26).

The numerical solution relies on two steps. First, in Section C.1, I use this assumption to derive a Hamilton-Jacobi-Bellman partial differential equation for the value function of the social planner V (26). Second, in Section C.2, I derive an approximating Markov chain, that is, a discretisation of the state space and a discrete Markov chain over it, which converges to the Hamilton-Jacobi-Bellman equations. This extends the techniques developed by Kushner and Dupuis (2001) and Kushner (2007) to a problem with recursive utility, such as the ones solved in Sections 3. I further provide a parallelisation technique based on Bierkens, Fearnhead and Roberts (2019).

C.1 Hamilton-Jacobi-Bellman equation

This section derives the Hamilton-Jacobi-Bellman (HJB) equation for the social planner problem and the emission game.

C.1.1 Hamilton-Jacobi-Bellman equation

The value function of the social planner V (26) at time t depends only on temperature T_t, log- $\mathrm{CO_2e}$ concentration m_t and output Y_t. This satisfies the HJB equation

$$\begin{aligned} -\partial_t V = \sup_{\chi,\alpha} \, & f(\chi Y, V) + \partial_m V \, (\gamma_t^{\mathrm{np}} - \alpha) + \partial_m^2 V \, \frac{\sigma_m^2}{2} \\ & + \partial_Y V (\varrho + \phi(\chi) - d(T, m) - A_t \beta_t(\varepsilon(\alpha))) Y + \partial_Y^2 V \, \frac{\sigma_Y^2}{2} Y^2 \\ & + \partial_T V \, \frac{r(T) + g(m)}{\epsilon} + \partial_T^2 V \, \frac{(\sigma_T/\epsilon)^2}{2}. \end{aligned} \tag{66}$$

Now, I derive an auxiliary, lower dimensional HJB equation using the ansatz

$$V_t(T, m, Y) \equiv \frac{Y^{1-\theta}}{1-\theta} F(T, m). \tag{67}$$

First, I compute the Epstein-Zin aggregator at the ansatz $V \equiv F \frac{Y^{1-\theta}}{1-\theta}$. Recall that the Epstein-Zin aggregator is given by

$$f(\chi Y, V) \coloneqq \rho \frac{(1-\theta) \, V}{1 - 1/\psi} \left(\left(\frac{\chi Y}{((1-\theta)V)^{\frac{1}{1-\theta}}} \right)^{1-1/\psi} - 1 \right). \tag{68}$$

Notice that

$$(1-\theta)V = F Y^{1-\theta} \tag{69}$$

$$((1-\theta)V)^{\frac{1}{1-\theta}} = F^{\frac{1}{1-\theta}} Y. \tag{70}$$

Plugging back into the aggregator we obtain

$$f(\chi Y, V) = Y^{1-\theta} \, \frac{\rho F}{1 - 1/\psi} \left(\left(\frac{\chi}{F^{\frac{1}{1-\theta}}} \right)^{1-1/\psi} - 1 \right). \tag{71}$$

Hence, f is homogenous of degree $1-\theta$ in Y. Second, we can evaluate the derivates of the value function at our ansatz

$$
\begin{aligned}
\partial_Y V &= Y^{-\theta}F, \qquad \partial_Y^2 V = -\theta Y^{-\theta-1}F,\\
\partial_T V &= \frac{Y^{1-\theta}}{1-\theta}\,\partial_T F,\ \partial_T^2 V = \frac{Y^{1-\theta}}{1-\theta}\,\partial_T^2 F,\\
\partial_m V &= \frac{Y^{1-\theta}}{1-\theta}\,\partial_m F,\ \partial_m^2 V = \frac{Y^{1-\theta}}{1-\theta}\,\partial_m^2 F,\\
\partial_t V &= \frac{Y^{1-\theta}}{1-\theta}\,\partial_t F.
\end{aligned}
$$

Plugging this into the HJB equation we obtain

$$
\begin{aligned}
-\frac{Y^{1-\theta}}{1-\theta}\partial_t F = \sup_{\chi,\alpha} \Bigg\{ & Y^{1-\theta}\frac{\rho F}{1-1/\psi}\left(\left(\frac{\chi}{F^{1/(1-\theta)}}\right)^{1-1/\psi}-1\right)\\
&+\frac{Y^{1-\theta}}{1-\theta}\partial_m F(\gamma_t^{\mathrm{np}}-\alpha)+\frac{Y^{1-\theta}}{1-\theta}\partial_m^2 F\frac{\sigma_m^2}{2}\\
&+Y^{-\theta}F\left(\varrho+\phi(\chi)-d(T,m)-A_t\beta_t(\varepsilon(\alpha))\right)Y\\
&-\theta Y^{-\theta-1}F\frac{\sigma_Y^2}{2}Y^2\\
&+\frac{Y^{1-\theta}}{1-\theta}\partial_T F\frac{r(T)+g(m)}{\epsilon}+\frac{Y^{1-\theta}}{1-\theta}\partial_T^2 F\frac{(\sigma_T/\epsilon)^2}{2}\Bigg\}
\end{aligned}
\tag{72}
$$

Dividing both sides by $\frac{Y^{1-\theta}}{1-\theta}$, we obtain

$$
\begin{aligned}
-\partial_t F = \inf_{\chi,\alpha} \Bigg\{ & (1-\theta)\frac{\rho F}{1-1/\psi}\left(\left(\frac{\chi}{F^{1/(1-\theta)}}\right)^{1-1/\psi}-1\right)\\
&+(1-\theta)F\left(\varrho+\phi(\chi)-d(T,m)-A_t\beta_t(\varepsilon(\alpha))-\theta\frac{\sigma_Y^2}{2}\right)\\
&+\partial_m F(\gamma_t^{\mathrm{np}}-\alpha)+\partial_m^2 F\frac{\sigma_m^2}{2}\\
&+\partial_T F\frac{r(T)+g(m)}{\epsilon}+\partial_T^2 F\frac{(\sigma_T/\epsilon)^2}{2}\Bigg\}.
\end{aligned}
\tag{73}
$$

C.1.2 Unitary Elasticity of Intertemporal Substitution $\psi \to 1$

Using the fact that

$$
\begin{aligned}
\lim_{\psi\to 1}\frac{\rho F}{1-1/\psi}\left(\left(\frac{\chi}{F^{1/(1-\theta)}}\right)^{1-1/\psi}-1\right) &= \rho F\ln\left(\frac{\chi}{F^{1/(1-\theta)}}\right)\\
&= \rho F\left(\ln(\chi)-\frac{\ln(F)}{1-\theta}\right)
\end{aligned}
\tag{74}
$$

we obtain a simplified HJB

$$
\begin{aligned}
-\partial_t F = \inf_{\chi,\alpha} \Bigg\{ & (1-\theta)\,\rho F \ln(\chi) - \rho F \ln(F) \\
& + (1-\theta) F \left(\varrho + \phi(\chi) - d(T,m) - A_t \beta_t(\varepsilon(\alpha)) - \theta \frac{\sigma_Y^2}{2} \right) \\
& + \partial_m F (\gamma_t^{\text{np}} - \alpha) + \partial_m^2 F \frac{\sigma_m^2}{2} \\
& + \partial_T F \frac{r(T) + g(m)}{\epsilon} + \partial_T^2 F \frac{(\sigma_T/\epsilon)^2}{2} \Bigg\}.
\end{aligned} \tag{75}
$$

Introduce the change of variable $H := \log(F)$, such that

$$
\begin{aligned}
&\frac{\partial_t F}{F} = \partial_t H, \\
&\frac{\partial_T F}{F} = \partial_T H, \quad \frac{\partial_T^2 F}{F} = (\partial_T H)^2 + \partial_T^2 H, \\
&\frac{\partial_m F}{F} = \partial_m H, \quad \frac{\partial_m^2 F}{F} = (\partial_m H)^2 + \partial_m^2 H.
\end{aligned}
$$

Using the change of variables we can rewrite (75) as

$$
\begin{aligned}
\rho H - \partial_t H = \inf_{\chi,\alpha} \Bigg\{ & (1-\theta) \left(\rho \ln(\chi) + \varrho + \phi(\chi) - d(T,m) - A_t \omega_t \varepsilon_t(\alpha)^b - \theta \frac{\sigma_Y^2}{2} \right) \\
& + (\gamma_t^{\text{np}} - \alpha) \partial_m H + \frac{\sigma_m^2}{2} \left((\partial_m H)^2 + \partial_m^2 H \right) \\
& + \frac{r(T) + g(m)}{\epsilon} \partial_T H + \frac{(\sigma_T/\epsilon)^2}{2} \left((\partial_T H)^2 + \partial_T^2 H \right) \Bigg\}.
\end{aligned} \tag{76}
$$

C.1.3 First Order Conditions

Under unitary elasticity of intertemporal substitution $\psi \to 1$, the optimal policy rates can be determined analytically

$$
\chi_t^* = \frac{\kappa A_t - 1 + \sqrt{(\kappa A_t - 1)^2 + 4\kappa\rho}}{2\kappa A_t} \tag{77}
$$

and

$$
\varepsilon_t^* = \left(- \frac{(\gamma_t^{np} + \delta_m(m))\, \partial_m H}{b A_t \omega_t (1-\theta)} \right)^{\frac{1}{b-1}}. \tag{78}
$$

Using this, (76) can be further simplified to

$$
\begin{aligned}
\rho H - \partial_t H = {} & (1-\theta)\left(\rho \ln(\chi_t^*) + \varrho + \phi(\chi_t^*) - d(T,m) - \theta \frac{\sigma_Y^2}{2}\right) \\
& + \frac{r(T) + g(m)}{\epsilon} \partial_T H + \gamma_t^{np} \partial_m H \\
& + \frac{\sigma_m^2}{2} \left(\partial_m H\right)^2 + \frac{\sigma_m^2}{2} \partial_m^2 H \\
& + (1-\theta)(b-1) A_t \omega_t \left(-\frac{\left(\gamma_t^{np} + \delta_m(m)\right) \partial_m H}{b A_t \omega_t (1-\theta)}\right)^{\frac{b}{b-1}} \\
& + \frac{(\sigma_T/\epsilon)^2}{2} \left(\left(\partial_T H\right)^2 + \partial_T^2 H\right).
\end{aligned}
\tag{79}
$$

C.2 Approximating Markov Chain

The objective is to solve for the function F by adapting the method proposed in Kushner and Dupuis (2001) and Kushner (2007). I discretise the state space of temperature T and $\mathrm{CO_2e}$ log-concentration m by a fixed grid of size parametrised by some small step size h. I discretise the time t by computing state dependent intervals $\Delta t(T, m)$. Then I define a Markov chain $\mathcal{M}$ over the discretised space. Finally, I compute a discretised value function $F^h : \mathcal{X} \to \mathcal{X}$ with the property that

$$
\sup_{x \in \mathcal{X}} |F^h(x) - F(x)| \to 0 \text{ as } h \to 0, \tag{80}
$$

for any compact subset $\mathcal{X} \subset [T^p, \infty) \times [m^p, \infty) \times [0, \infty)$. For more details on convergence, see Chapter 15 of Kushner and Dupuis (2001).

First, I fix a suitable subset of the state space

$$
\mathcal{X} := [T^p, T^p + \Delta T] \times [m^p, m^p + \Delta m]. \tag{81}
$$

In practice ΔT and Δm are chosen such that T in the upper-right corner of the grid, the drift $\mu(T^p + \Delta T, m^p + \Delta m) \equiv 0$. Then, given a step size $h > 0$ I define the grid over the space $\mathcal{X}$ by

$$
\begin{aligned}
\Omega_h(\mathcal{X}) = {} & (T^p, T^p + h\Delta T, \ldots, T^p + (1-h)\Delta T, T^p + \Delta T) \times \\
& (m^p, m^p + h\Delta m, \ldots, m^p + (1-h)\Delta m, m^p + \Delta m).
\end{aligned}
\tag{82}
$$

Using the ansatz (67) I now define a discretised value function F_t^h over the grid such that

$F_t^h \to F$ as $h \to 0$. First, introduce the discrete-time Epstein-Zin aggregator

$$J^h(\chi, F) = \left(\left(1 - e^{-\rho\Delta t}\right) \chi^{1-\frac{1}{\psi}} + e^{-\rho\Delta t} \Big(\delta_y(\chi) F\Big)^{\frac{1-\frac{1}{\psi}}{1-\theta}} \right)^{\frac{1-\theta}{1-\frac{1}{\psi}}}, \tag{83}$$

where

$$\begin{aligned} \delta_y(\chi) &:= \mathbb{E}_t \left[\left(\frac{Y_{t+\Delta t}}{Y_t} \right)^{1-\theta} \right] \\ &= 1 + \Delta t(1-\theta) \left(\varrho + \phi(\chi) - d(T_t) - \frac{\theta}{2}\sigma_k^2 \right) + \mathbb{E}_t \mathcal{O}\left(\Delta t^{\frac{3}{2}} \right). \end{aligned} \tag{84}$$

As the aggregator J^h satisfies Assumptions 1 through 4 of Kraft and Seifried (2014, p. 534), we have uniform convergence of the discrete-time operator to the continuous analogue, that is,

$$\sup_{F \in \mathcal{F}, \chi \in \mathcal{C}} \left| \frac{J^h(\chi, F) - J(\chi, F)}{h} - g(\chi, F) \right| \to 0 \text{ as } h \to 0, \tag{85}$$

for each compact $\mathcal{F} \subset \mathbb{R}$ and $\mathcal{C} \subseteq [0, 1]$. The discretised value function satisfies the recursion

$$F^h(T_t, m_t) = \min_{\chi, \alpha} J^h \left(\chi, \mathbb{E}_{t, \mathcal{M}(\alpha)} F_{t+\Delta t}(T_t, m_t) \right), \tag{86}$$

where $\mathbb{E}_{t,\mathcal{M}(\alpha)}$ is the expectation with respect to the Markov chain $\mathcal{M}(\alpha)$ over the grid $\Omega_h(\mathcal{X})$. The Markov chain $\mathcal{M}(\alpha)$, parameterised by h, is constructed as follows. Introduce the normalising factor

$$Q_t(T, m, \alpha) := \left(\frac{\sigma_T}{\epsilon \Delta T} \right)^2 + \left(\frac{\sigma_m}{\Delta m} \right)^2 + h \left| \frac{r(T) + g(m)}{\epsilon \Delta T} \right| + h \left| \frac{\gamma_t^{\text{np}} - \alpha}{\Delta m} \right|. \tag{87}$$

Then the probabilities of moving from a point $(T, m) \in \Omega_h(\mathcal{X})$ of the grid to an adjacent point are given by

$$p(T \pm h\Delta T, m \mid T, m) \propto \frac{1}{2} \left(\frac{\sigma_T}{\epsilon \Delta T} \right)^2 + h \left(\frac{r(T) + g(m)}{\epsilon \Delta T} \right)^{\pm} \text{ and} \tag{88}$$

$$p(T, m \pm h\Delta m \mid T, m) \propto \frac{1}{2} \left(\frac{\sigma_m}{\Delta m} \right)^2 + h \left(\frac{\gamma_t^{\text{np}} - \alpha}{\Delta m} \right)^{\pm} \tag{89}$$

where $(\cdot)^+ := \max\{\cdot, 0\}$ and $(\cdot)^- := -\min\{\cdot, 0\}$. One can check that this is a well-defined probability measure. Finally, the time step is given by

$$\Delta t = h^2 / Q_t(T, m, \alpha), \tag{90}$$

which satisfies $\Delta t \to 0$ as $h \to 0$.

The Markov chain defined above allows us to derive $F^h_t(T_t, m_t)$ from the subsequent $F^h_{t+\Delta t}(T_{t+\Delta t}, m_{t+\Delta t})$. This requires a terminal condition

$$\bar{F}^h(T_\tau, m_\tau) \coloneqq F^h_\tau(T_\tau, m_\tau). \tag{91}$$

To derive this, assume that at some point in a far future $\tau \gg 0$, the abatement is free and all emissions are abated, $\gamma^{\mathrm{np}} \equiv \alpha$ for the social planner such that $\mathrm{d}m \equiv \sigma_m \mathrm{d}W_m$. Then I construct an equivalent, control independent, Markov chain $\bar{\mathcal{M}}$ as above for the steady state value function

$$\bar{F}^h(T, m) = \min_{\chi} J^h\left(\mathbb{E}_{t,\bar{\mathcal{M}}} \bar{F}(T, m); \chi\right). \tag{92}$$

This is now a fixed point equation for $\bar{F}$ which can be solved by value or policy function iteration. The controlled drift

$$\begin{pmatrix} \mu(T_t, m_t) & \alpha_t - \gamma^{\mathrm{np}}_t \end{pmatrix} \tag{93}$$

is continuous in the state vector. Furthermore, the aggregator J^h is bounded and continuous. This implies that the problem at hand satisfies Assumptions 10.1.1 and 10.1.2 for (weak) convergence given in Kushner and Dupuis (2001, p. 269). Hence, we have convergence of the discretised value function to the continuous analogue, $\bar{F}^h \to \bar{F}$. The steady state value function $\bar{F}$ serves then as a continuous and bounded terminal condition of the finite horizon problem (86). Which, in turn, implies (weak) convergence of $F^h_t \to F$ as $h \to 0$.

C.2.1 Convergence of Numerical Method

The convergence of the steady state HJB equation (92) for $\bar{F}$ is achieved via iterated application of the right-hand side. As J is contracting, repeated application is guaranteed to converge to the unique fixed point $\bar{F}^h$. Convergence is determined when the absolute and relative tolerance both fall below than 1×10^{-7} and 1×10^{-8} respectively. The convergence values are shown below (Table 11). To check the robustness of the convergence procedure, I also computed the terminal value function and policy on a denser grid with step size by $h/2$ and check the maximum absolute error on the coarser grid. These are reported in the fourth column.

Furthermore, I employ a verification test based on Oberkampf and Roy (2010), as in Cai and Lontzek (2019). I consider a determnistic model $\sigma_m = \sigma_T = 0$, such that the HJB equation

T^c	Iterations	Abs. Tolerance	Policy error with $h/2$
Linear	8950	9.96×10^{-8}	1.345×10^{-3}
$2°$	9441	9.9972×10^{-8}	2.268×10^{-3}
$3°$	9449	9.9973×10^{-8}	2.593×10^{-3}
$4°$	9440	9.9972×10^{-8}	2.113×10^{-3}

Table 11: Convergence metrics for different critical thresholds with $h = 4 \times 10^{-3}$. The policy error is the absolute difference in policy for a denser grid $h/2$.

(76) can be solved via a upwind-downwind finite-difference scheme. The solution of the latter is then compared to the solution of the approximating Markov Chain method described above. The error between the two methods is displayed in Table 12.

T^c	Abs. Error
Linear	4.2×10^{-4}
$2°$	7.1×10^{-3}
$3°$	8.2×10^{-3}
$4°$	8.7×10^{-3}

Table 12: Absolute error in the value function for H computed via an approximating Markov chain and an upwind-downwind finite-difference scheme.

C.3 Parallelisation

When computing the backward recurrence (86), conditional on a given strategy α, each grid point $X_i \in \Omega^h(\mathcal{X})$ is assigned a different time step $\Delta t(X_i; \alpha)$, which depends on the curvature of the drift at that state. To parallelise the computation, I leverage the ZigZag algorithm by Bierkens, Fearnhead and Roberts (2019). Given the value function F^h_t, for a step back $t - \min_i \Delta t(X_i; \alpha)$, I construct a directed graph among grid points $\mathcal{X}$ where an edge $X_i \to X_j$ is drawn if there is a positive probability of transitioning from X_i to X_j under the Markov Chain $\mathcal{M}$. This allows to obtain, at each point in time t, sets of points $\mathcal{C}_t \subseteq \mathcal{X}$ which are independent and over which it is possible to parallelise. The parallelisation has been conducted on Snellius, the national high performance computer of the Netherlands. The algorithm is written in the `Julia` programming language and relies on `Optim.jl` developed by Mogensen and Riseth (2018) and the differential equation suit developed by Rackauckas and Nie (2017).

D Robust Policy

D.1 Computation of Robust Policy φ^r

Each function φ in the restricted policy space Φ (31) is determined by a weight vector $w \in \Delta^{|\mathcal{T}|}$, where $\Delta^{|\mathcal{T}|}$ is simplex over the space of critical thresholds $\mathcal{T}$ (30). Let

$$P_{\mathcal{T}} := \begin{pmatrix} \varphi^{\underline{T}^c} & \varphi^{\underline{T}^c + \Delta T^c} & \varphi^{\underline{T}^c + 2\Delta T^c} & \dots & \varphi^{\overline{T}^c} & \overline{\varphi} \end{pmatrix}' \tag{94}$$

be the vector of optimal policies over the critical threshold space $\mathcal{T}$. The min-max regret problem (32) can then be equivalently written as finding the minimiser w^r of

$$\min_{w \in \Delta^{|\mathcal{T}|}} \left\{ \max_{T^c \in \mathcal{T}} \mathcal{R}\left(w \cdot P_{\mathcal{T}}, T^c; t, X_t\right) \right\}. \tag{95}$$

We are interested in the robust policy at the initial condition. I denote the initial regret as $\mathcal{R}_0(w, T^c) := \mathcal{R}\left(w \cdot P_{\mathcal{T}}, T^c; 0, X_0\right)$.

As $\mathcal{T}$ is a finite space, this problem admits an "epigraph" reformulation[8], that is, one can equivalently solve the constrained minimisation problem

$$\min_{y \in \mathbb{R}, w \in \Delta^{|\mathcal{T}|}} y \quad \text{s.t. } y \geq \mathcal{R}_0(w, T^c)\ \forall T^c \in \mathcal{T}. \tag{96}$$

The constrained optimisation problem is solved using the conservative convex separable approximation (CCSA) algorithm introduced in Svanberg (2002) and implemented in `nlopt` by Johnson (2007).

I set $\underline{T}^c = 2°$, $\overline{T}^c = 5°$ and $\Delta T^c = 0.025°$. To check the robustness of this choice I compute the regret contribution of decreasing the lower threshold $\underline{T}^c < 2°$ and increasing the upper threshold $\overline{T}^c > 5°$. The maximum regret contribution is $\max_{T^c < 2° \text{ and } T^c > 5°} R_0(\varphi^r, T^c) < 1 \times 10^{-4}$. On the one hand, increasing the upper bound contributes little to the regret because the optimal policy under a linear climate $\overline{\varphi}$ stabilises the temperature at a sufficiently low level that the probability of crossing a critical threshold of $5°$ is small. On the other hand, reducing the lower bound contributes little to the regret as the optimal policy under a imminent tipping point $\varphi^{T^c < 2°}$ has a low probability of preventing the tipping point from occurring, hence, the regret associated with not implementing such policy is low.

[8] I thank Steven G. Johnson for pointing this out.